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Review

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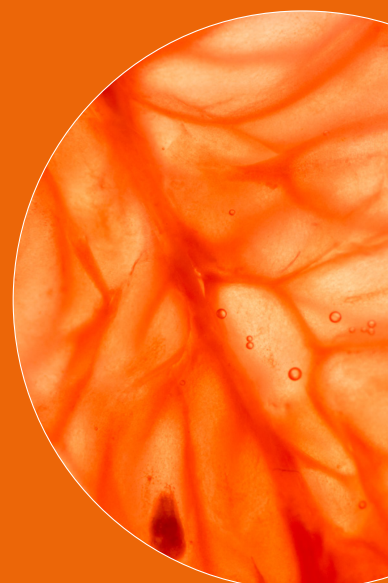
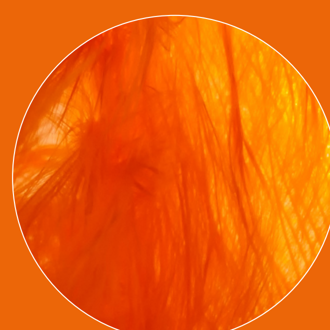


FIVE TECHNOLOGY TRENDS
SHAPING ICT INNOVATION

NB-IoT: SUSTAINABLE TECHNOLOGY



CLOUD ROBOTICS
ENABLED BY 5G

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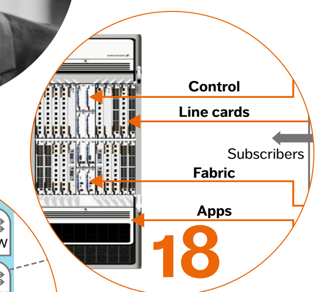
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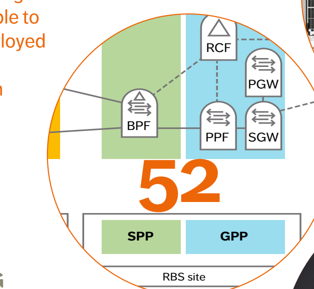
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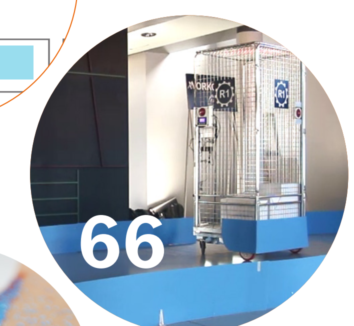
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THE **IoT**, SPEED, AND **DEEP** COLLABORATION

"TECHNOLOGY DEVELOPMENTS IN ONE INDUSTRY CAN RAPIDLY PROVIDE BENEFIT TO OTHER MARKET SECTORS."

■ **DEVELOPMENTS IN TECHNOLOGY** have contributed to the launch of innumerable products and solutions designed to make our lives easier. A smartphone can help us find out where we are, conduct research, watch a favorite movie, make a video call to a friend, read a magazine article, find the nearest restaurant, book tickets, or send someone a picture – even while flying. Advancements in technology have helped to reunite refugees with their families, and combat terrorism. They have brought medical care into the living room, tearing down the obstacles and boundaries of traditional business models. The benefits brought about through research not only apply to telecoms, but to all modern industries. But the significant change that has resulted in the rapid deployment of innovation is the way industries collaborate today, with technology developments in one industry rapidly providing benefits to other market sectors.

According to the June 2016 Ericsson Mobility Report, connected IoT devices will outnumber mobile phones by 2018. This forecast reminds me of the milestone we witnessed at the end of 2009, when data traffic surpassed voice traffic in mobile networks. Since then, voice traffic has remained more or less constant, yet data traffic has continued to demonstrate strong growth.

The smartphone revolution that followed brought with it a massive amount of network adaptation, so that networks built for voice could be transformed into data carriers. Today, the IoT presents a similar need for change. With features like billing, in-app purchasing, and video streaming, most mobile-broadband networks have been designed to support the traffic generated by typical subscribers. But the IoT, with its wide range of applications, needs customized connectivity: in other words, connectivity that suits each application in terms of cost, reach, bandwidth, and latency. The bottom line is cost. It simply doesn't make economic sense to use broadband networks and valuable spectrum for applications that transmit just a few kbs of data now and then. And so the need has arisen for a radio-access technology that can meet the connectivity requirements for massive MTC applications. This technology is narrowband IoT (NB-IoT), a solution designed to be deployed in the GSM spectrum, within an LTE carrier, or in an LTE or WCDMA guard band.

Robotics is one of the clear winners when it comes to the cross-fertilization of developments from different disciplines. The field of robotics brings materials, communication, and manufacturing together. The result: highly sophisticated production processes that are adapted to just-in-time methodologies, zero-waste policies, minimum use of raw materials, and low energy consumption. 5G is a key ingredient that will help to make the robotics industry mass market and affordable. By providing the connectivity that will support even the most demanding applications, 5G will enable system intelligence to be transferred to a cloud where computational capacity is greatest, and put simplified – more affordable – robots on the ground. And once people start to rely on robots – just as they depend on their smartphones – to carry out the practical tasks of daily life, we can expect this sector to boom.

Security will continue to present new challenges. But as security issues continue to dominate headline news, developments are shifting from fire-fighter mode to prevention. You can read more of my thoughts on the shift in technology in the Tech Trends section of this issue of Ericsson Technology Review. Apart from security, my trends for the coming year include: the ability of the cloud to spread intelligence, self-managing devices, communication beyond sight and sound, and the influence of other sectors.

If I was to suggest one takeaway from all of the articles included in this issue, I would say it is speed. Device processing is getting faster, data speeds are constantly increasing, radio speeds are approaching those of fiber, more people are becoming subscribers, more things are becoming connected, more applications are running constantly. Developers of new technologies are working hard to enhance responsiveness by reducing latency, a key performance parameter. The capability to determine what functions can be virtualized to maximize ideal placement in the network and ensure low latency is one of the primary driving factors behind a proposed split of radio-access architecture, which is detailed in the article 4G/5G RAN architecture: how a split can make the difference.

As always, I hope you find our stories relevant and inspiring. All of our content is available at *www.ericsson.com/ericsson-technology-review*, through the Ericsson Technology Insights app, and on SlideShare.

ULF EWALDSSON
SENIOR VICE PRESIDENT, GROUP CTO AND HEAD OF GROUP FUNCTION TECHNOLOGY

NB-IoT:

A SUSTAINABLE TECHNOLOGY FOR **CONNECTING BILLIONS OF DEVICES**

Under the umbrella of 3GPP, radio-access technologies for mobile broadband have evolved effectively to provide connectivity to billions of subscribers and things. Within this ecosystem, the standardization of a radio technology for massive MTC applications – narrowband IoT (NB-IoT) – is also evolving. The aim is for this technology to provide cost-effective connectivity to billions of IoT devices, supporting low power consumption, the use of low-cost devices, and provision of excellent coverage – all rolled out as software on top of existing LTE infrastructure. The design of NB-IoT mimics that of LTE, facilitating radio network evolution and efficient coexistence with MBB, reducing time to market, and reaping the benefits of standardization and economies of scale.

SARA LANDSTRÖM
JOAKIM BERGSTRÖM
ERIK WESTERBERG
DAVID HAMMARWALL

THE BEST WAY to provide MTC applications with cost-effective connectivity is to design the radio-access network accordingly. What is needed is a radio-access network that minimizes battery usage, covers a wide area, and functions with simplified low-cost devices while efficiently matching the varying spectrum allocations of operators. 3GPP release 13 specifications includes the NB-IoT feature, with a large degree of deployment flexibility to maximize migration possibilities and allow the technology to be deployed in GSM spectrum, in an LTE carrier, or in a WCDMA or LTE guard band.

The IoT embeds a broad range of MTC applications, and among the different types, it is widely accepted that massive MTC will be the first to take off. This segment includes applications like smart metering, agriculture and real estate monitoring, as well as various types of tracking and fleet management. Often referred to as low power wide area (LPWA), networks that provide connectivity to massive MTC applications require a radio-access technology that can deliver widespread coverage, capacity, and low power consumption.

Massive MTC devices typically send small amounts of data, and tend to be placed in signal-challenged locations like basements and remote rural areas. Due to the sheer numbers of devices deployed in typical scenarios, per-device and life-cycle costs need to be kept to a minimum, and measures that promote battery longevity are essential for ensuring the overall cost-effectiveness of the system.

The coverage and throughput needs for massive MTC applications are quite different from those of MBB. The need to support high bitrates, for example, applies to MBB scenarios, but seldom to massive MTC. The precise nature of massive MTC allows for a significant degree of optimization in the design of radio access.

Standardization of NB-IoT began in 2014 with a 3GPP study. The objective of this study was to determine the requirements for massive MTC, to choose an evaluation methodology, and to investigate whether proposed radio-access designs could meet the set requirements. This study led to work on the specification of NB-IoT [1], with a number of design targets – as illustrated in *Figure 1*.

In addition to the design targets, extensive deployment flexibility and use of industry competence to meet time-to-market requirements

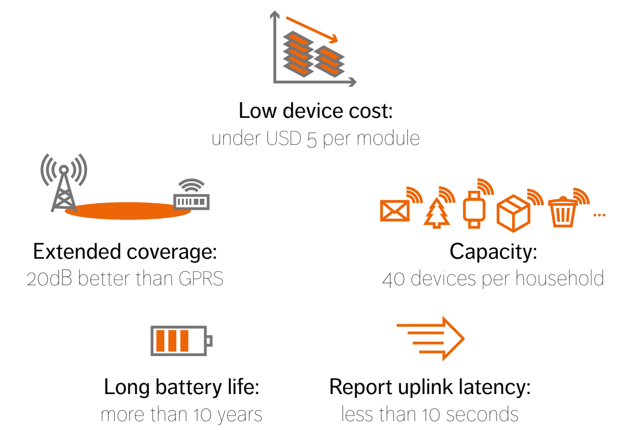

Figure 1
IoT design targets

Terms and abbreviations

CS – circuit-switched | **DL** – downlink | **DRX** – discontinuous reception | **eDRX** – extended DRX | **eMBMS** – evolved multimedia broadcast multicast service | **eMTC** – enhanced machine-type communications | **EPC** – Evolved Packet Core | **E-UTRA** – Evolved Universal Terrestrial Radio Access | **IoT** – Internet of Things | **LPWA** – low power wide area | **MAC** – media/medium access control | **MBB** – mobile broadband | **MTC** – machine-type communications | **NB-IoT** – narrowband Internet of Things | **OFDMA** – Orthogonal Frequency-Division Multiple Access | **PA** – power amplifier | **PRB** – physical resource block | **PSM** – power save mode | **RF** – radio frequency | **RLC** – Radio Link Control | **RRC** – Radio Resource Control | **SC-FDMA** – single-carrier frequency-division multiple access | **TCO** – total cost of ownership | **UE** – user equipment | **UL** – uplink have been included as key considerations in the specification of NB-IoT. To future-proof the technology, its design exploits synergies with LTE by reusing the higher layers (RLC, MAC, and RRC), for example, and by aligning numerology (the foundation of the physical layer) in both the uplink and downlink. However, the access procedures and control channels for NB-IoT are new.

Prior to NB-IOT specification, work had already begun on the design of another radio access for massive MTC to support Cat-M1 – a new UE category. With completion also targeted for release 13, the resulting standardization work item – eMTC – covers bitrates, for example, ranging from hundreds of kbps to 1Mbps. These requirements are broader than NB-IoT which has been streamlined for applications with widely varying deployment characteristics, lower data rates, and operation with simplified low-cost devices.

With a carrier bandwidth of just 200kHz (the equivalent of a GSM carrier), an NB-IoT carrier can be deployed within an LTE carrier, or in an LTE or WCDMA guard band*. The link budget of NB-IoT has a 20dB improvement over LTE Advanced. In the uplink, the specification of NB-IoT allows for many devices to send small amounts of data in parallel.

*Guard band is a thin band of spectrum between radio bands that is used to prevent interference.

Release 13 not only includes standards for eMTC and NB-IoT, it also contains important refinements, such as extended discontinuous reception (eDRX) and power save mode (PSM). PSM was completed in release 12 to ensure battery longevity, and is complemented by eDRX for use cases involving devices that need to receive data more frequently.

Deployment flexibility and migration scenarios

As a finite and scarce natural resource, spectrum needs to be used as efficiently as possible. And so technologies that use spectrum tend to be designed to minimize usage. To achieve spectrum efficiency, NB-IoT has been designed with a number of deployment options for GSM, WCDMA, or LTE spectrum, which are illustrated in *Figure 2*.

- standalone – replacing a GSM carrier with an NB-IoT carrier
- in-band – through flexible use of part of an LTE carrier
- guard band – either in WCDMA or LTE

Starting with standalone

The standalone deployment is a good option for WCDMA or LTE networks running in parallel with GSM. By steering some GSM/GPRS traffic to the WCDMA or LTE network, one or more of the GSM carriers can be used to carry IoT traffic. As GSM operates mainly in the 900MHz and 1,800MHz bands (spectrum that is present in all markets), this approach accelerates time to market, and maximizes the benefits of a global-scale infrastructure.

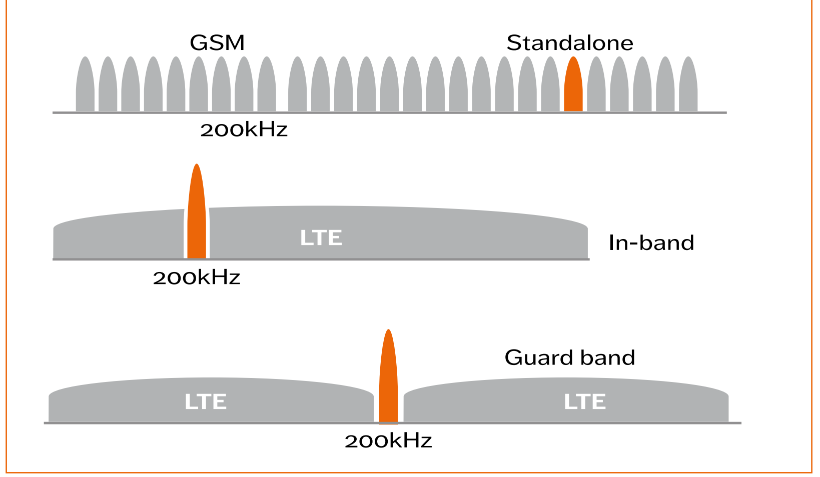

Figure 2
Spectrum usage deployment options

Migration to in-band

When the timing is right, GSM spectrum will be refarmed for use by more demanding MBB traffic. Refarming spectrum for use by LTE is a straightforward process, even when NB-IoT carriers exist in the GSM spectrum because refarming does not impact NB-IoT devices, and any NB-IoT carriers in GSM will continue to operate within the LTE carrier after migration. Such a future-proof setup is possible, as the standalone and in-band modes use the same numerology as LTE, and RF requirements are set to match the different deployments, so all devices are guaranteed to support in-band operation at the time of migration.

In-band: best option for LTE

For operators with mainly LTE spectrum available, the LTE in-band option provides the most spectrum- and cost-efficient deployment of NB-IoT. More than anything else, this particular option sets NB-IoT apart from any other LPWA technology.

An NB-IoT carrier is a self-contained network element that uses a single physical resource block (PRB). For in-band deployments with no IoT traffic present, the PRB can be used by LTE for other purposes, as the infrastructure and spectrum usage of LTE and NB-IoT are fully integrated. The base station scheduler multiplexes NB-IoT and LTE traffic onto the same spectrum, which minimizes the total cost of operation for MTC, which essentially scales with the volume of MTC traffic. In terms of capacity, the capability of a single NB-IoT carrier is quite significant – evaluations have shown that a standard deployment can support a deployment density of 200,000 NB-IoT devices within a cell – for an activity level corresponding to common use cases. Naturally, more NB-IoT carriers can be added if more capacity is needed.

Using guard band spectrum

A third alternative is to deploy NB-IoT in a guard band, and here, the focus is on the use of such bands in LTE. To operate in a guard band without causing interference, NB-IoT and LTE need to coexist. In contrast to other LPWA technologies, the physical NB-IoT layers have been designed with the requirements of in-LTE-guard-band coexistence specifically taken into consideration. Again, like LTE, NB-IoT uses OFDMA in the downlink and SC-FDMA in the uplink.

The design of NB-IoT has fully adopted LTE numerology, using 15kHz subcarriers in the uplink and downlink, with an additional option for 3.75kHz subcarriers in the uplink to provide capacity in signal-strength-limited scenarios.

Long range and long battery life

The geographical area for which a mobile network can provide coverage depends on site density and link budget. Compared with GPRS, WCDMA and LTE, the link budget of NB-IoT has a 20dB margin, and use cases tend to operate with lower data rates.

So, not only can NB-IoT reuse the GSM, WCDMA, or LTE grid, the improved link budget enables it to reach IoT devices in signal-challenged locations such as basements, tunnels, and remote rural areas – places that cannot be reached using the network's voice and MBB services.

In technical terms, the coverage target of NB-IoT has a link budget of 164dB, whereas the current GPRS link budget is 144dB (TR 45.820 [2]), and LTE is 142.7 dB** (TR 36.888 [3]). The 20dB improvement corresponds to a sevenfold increase in coverage area for an open environment, or roughly the loss that occurs when a signal penetrates the outer wall of a building. Standardization activities in 3GPP have shown that NB-IoT meets the link budget target of 164dB, while simultaneously meeting the MTC application requirements for data rate, latency, and battery life.

The battery life of an MTC device depends to some extent on the technology used in the physical layer for transmitting and receiving data. However, longevity depends to a greater extent on how efficiently a device can utilize various idle and sleep

** The noise figure assumptions in 3GPP TS 36.888 [3] used in the link budget calculations are more conservative than in the corresponding link budget for GSM in 3GPP TR 45.820. Using the noise figure assumptions from TR 45.820, the LTE link budget becomes 142.7dB.

modes that allow large parts of the device to be powered down for extended periods. The NB-IoT specification addresses both the physical-layer technology and idling aspects of the system.

Like LTE, NB-IoT uses two main RRC protocol states: RRC_idle and RRC_connected. In RRC_idle, devices save power, and resources that would be used to send measurement reports and uplink reference signals are freed up. In RRC_connected, devices can receive or send data directly.

Discontinuous reception (DRX) is the process through which networks and devices negotiate when devices can sleep and can be applied in both RRC_idle and RRC_connected. For RRC_connected, the application of DRX reduces the number of measurement reports devices send and the number of times downlink control channels are monitored, leading to battery savings.

3GPP release 12 supports a maximum DRX cycle of 2.56 seconds, which will be extended to 10.24 seconds in release 13 (eDRX). However, any further lengthening of this period is as yet not feasible, as it would negatively impact a number of RAN functions including mobility and accuracy of the system information. In RRC_idle, devices track area updates and listen to paging messages. To set up a connection with an idle device, the network pages it. Power consumption is much lower for idle devices than for connected ones, as listening for pages does not need to be performed as often as monitoring the downlink control channel.

When PSM was introduced in release 12, it enabled devices in RRC_idle to enter a deep sleep in which pages are not listened for, nor are mobility-related measurements performed. Devices in PSM perform tracking area updates after which they directly listen for pages before sleeping again. PSM and eDRX complement each other and can support battery lifetimes in excess of 10 years for different reachability requirements, transmission frequencies of different applications, and mobility.

The range of solutions designed to extend battery lifetimes need to be balanced against requirements for reachability, transmission frequency of different applications, and mobility. These relations are illustrated in *Figure 3*.

Superior capacity design

To meet capacity requirements, NB-IoT needs to multiplex many devices simultaneously, and provide connectivity in an efficient manner for all of them irrespective of coverage quality. As a result, the design of NB-IoT supports a range of data rates.

The achievable data rate depends on the channel quality (signal to noise ratio), and the quantity of allocated resources (bandwidth). In the downlink, all devices share the same power budget and several may simultaneously receive base-station transmissions. In the uplink, however, each device has its own power budget, and this can be used to advantage by multiplexing the traffic generated by several devices, as their combined power is greater than that of a single device.

In many locations, NB-IoT devices will be limited by signal strength rather than transmission bandwidth. Such devices can concentrate their transmission energy to a narrower bandwidth without loss of performance, which frees up bandwidth for others. The possibility of allocating small amounts of bandwidth to specific devices increases system capacity without loss of performance.

To enable such small bandwidth allocations, NB-IoT uses tones or subcarriers instead of resource blocks. The subcarrier bandwidth for NB-IoT is 15kHz, compared with a resource block, which has an effective bandwidth of 180kHz. Each device is scheduled on one or more subcarriers in the uplink, and devices can be packed even closer together by decreasing the subcarrier spacing to 3.75kHz. Doing so, however, results in differing numerology for LTE and NB-IoT, and some resources will need to be allocated to avoid interference between the 3.75kHz and 15kHz subcarriers instead of utilizing them for traffic, which may lead to performance losses.

For scenarios that include devices in both good and bad coverage areas, it is possible to increase the data rate by adding more bandwidth. In the uplink, data rates can be increased up to 12 times by allocating devices with a multi-tone or multi-subcarrier rather than a single tone, for example. This approach improves capacity for scenarios where many devices have good coverage, as data transfer completes quickly. Good coverage is typical

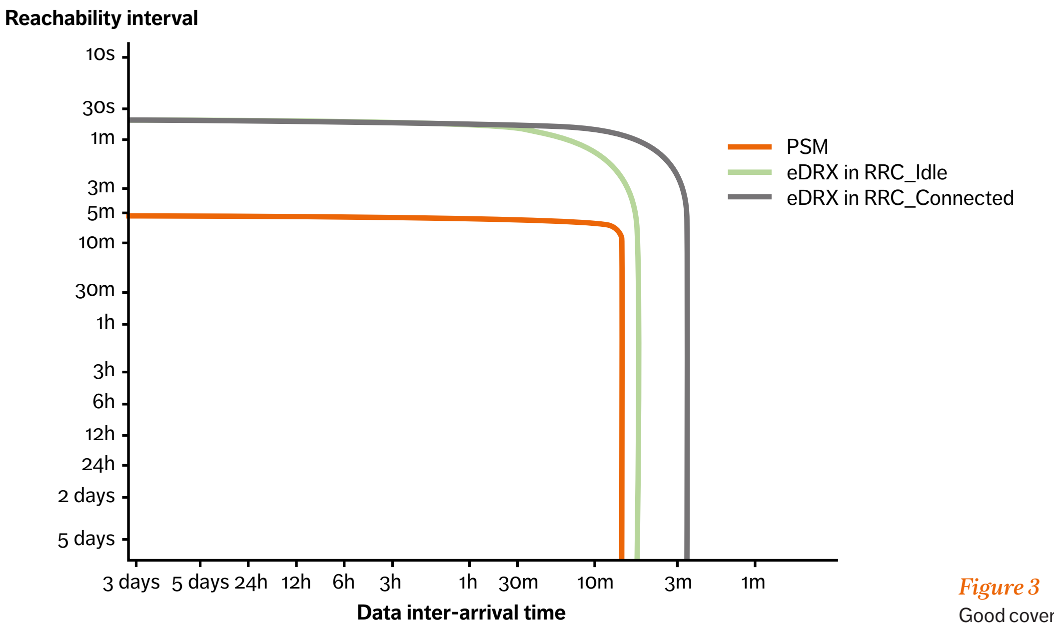

Figure 3
Good coverage

when NB-IoT is rolled out on a dense grid and/or when most devices are within the original LTE cell coverage area.

Data rate is a significant factor when trying to achieve the best design for NB-IoT, as it affects both latency and power consumption. *Table 1* shows the uplink latency values for a device to connect and transmit data. The data rates for worst-case coverage (+20dB) are lower than those for MBB at the cell edge (0dB), and latency increases from 1.6 to 7.6 seconds. The uplink data rate is the main cause of this degradation, yet even for worst-case scenarios, NB-IoT uplink latency is still under the 10-second design target. When it comes to power consumption, the dominating factor is the speed at which devices transmit data, which increases in line with accelerating data rates.

NB-IoT has been designed with good multiplexing and adaptable data rates and so it will be able to meet predicted capacity requirements. The capacity requirements target in 3GPP TR 45.820 [1] has been set to 40 devices per household, based on assumptions for London, which correspond to 52,500 devices per cell. Simulations show support for 200,000 devices per cell – four times the set target.

Device aspects

Affordable modems are a key element of large-scale sensor deployment, so that processes such as temperature or water meter reporting can be optimized. At the same time, the data rate and latency requirements of such sensor-heavy applications tend to be relatively modest: a characteristic that can be used to advantage to reduce solution complexity – and cost.

NB-IoT devices support reduced peak physical layer data rates: in the range of 100-200kbps or significantly lower for single-tone devices. To facilitate low-complexity decoding in devices, turbo codes are replaced with convolutional codes for downlink transmissions, and limits are placed on maximum transport block size – which is 680 bits for DL and not greater than 1000 bits for UL.

The performance requirements set for NB-IoT make it possible to employ a single receiver antenna (two are needed for LTE MBB). As a result, the radio and baseband demodulator parts of the device need only a single receiver chain. By operating NB-IoT devices half duplex so that they cannot be scheduled to send and receive data simultaneously, the duplex filter in the device can be replaced by a simple switch, and a only single local oscillator for frequency generation is required. These optimizations reduce cost and power consumption.

At 200kHz, the bandwidth of NB-IoT is substantially narrower than other access technologies. LTE bandwidths, for example, range from 1.4MHz to 20MHz. The benefit of a narrowband technology lies in the reduced complexity of analog-to-digital (A/D) and digital-to-analog (D/A) conversion, buffering, and channel estimation – all of which bring benefits in terms of power consumption.

NB-IoT brings about a significant design change in terms of the placement of the device's power amplifier (PA). Integrating this element directly onto the chip, instead of it being an external component, enables single-chip modem implementations – which are cheaper.

Table 1
Maximum uplink latency for a device on the MBB cell border (+0dB) and beyond (+10dB and +20dB)

Coverage	Sync	MIB	PRACH	RAmsg2-4	ULgrant	ULdata	Ack	ULdata	TOTAL
+0dB	340	151	324	622	48	39	41	39	1,604
+10dB	340	151	688	708	45	553	47	553	3,085
+20dB	520	631	1,440	1,060	49	1,923	77	1,923	7,623

Duration (ms)

Reuse of existing technology

The design of NB-IoT radio access reuses a number of LTE design principles and has the backing of the traditional cellular-network and chipset vendors that made MBB a success. NB-IoT employs the same design principles as LTE (E-UTRA), although it uses a separate new carrier, new channels, and random access procedures to meet the target requirements of IoT use cases – such as improved coverage, lower battery consumption and operation in narrow spectrum. Constructing NB-IoT in this way takes advantage of LTE's well-established global reach, economies of scale, and industry-leading ecosystem.

The NB-IoT downlink is based on OFDMA and maintains the same subcarrier spacing, OFDM symbol duration, slot format, slot duration, and subframe duration as LTE. As a result, NB-IoT can provide both in-band and guard band deployment without causing interference between its carriers and those used by LTE for MBB, making NB-IoT a well integrated IoT solution for LTE-focused operators in addition to Cat-M1.

Use of the same upper layers is yet another similarity between LTE and NB-IoT, with some optimizations to support operation with low-cost devices. For example, as a single technology solution,

> DATA RATE IS A SIGNIFICANT FACTOR WHEN TRYING TO ACHIEVE THE BEST DESIGN FOR NB-IOT, AS IT AFFECTS BOTH LATENCY AND POWER CONSUMPTION

NB-IoT does not support dual connectivity; and devices do not support switching between access technologies (GSM, WCDMA, or Wi-Fi) in active mode. Support for CS voice services has also been removed. These scope savings result in a much lower requirement for memory capacity for NB-IoT devices compared with even the most rudimentary MBB LTE ones.

NB-IoT uses an S1-based connection between the radio network and the EPC. The connection to the EPC provides NB-IoT devices with support for roaming and flexible charging, meaning that devices can be installed anywhere and can function globally. The ambition is to enable certain classes of devices – like smoke detectors – to be handled with priority to ensure that emergency-situation data can be prioritized if the network is congested.

Existing 3GPP architecture provides a global, highly automated connectivity management solution that is needed for large-scale IoT deployments. NB-IoT and LTE use the same O&M framework, running as a single network carrying MBB and MTC traffic, which reduces operational costs in areas like provisioning, monitoring, billing, and device management. Similar to present LTE networks, NB-IoT supports state-of-the-art 3GPP security, with authentication, signaling protection, and data encryption.

LTE features that already exist, like cell-ID-based positioning, are straightforward enough for NB-IoT to inherit. By aligning with LTE evolution, NB-IoT could support existing features and future functionality designed for the entire cellular ecosystem, including MBB as well as IoT use cases.

NB-IoT: the advantages of being part of 3GPP

- use of the LTE ecosystem, leading to fast development, economies of scale, and global roaming
- can be deployed as a simple addition of new software to existing LTE infrastructure
- a management framework exists, enabling large-scale deployments
- framework includes state-of-the-art security
- future feature growth for MBB and NB-IoT use cases

The broadcast feature eMBMS enables a large number of devices to be updated simultaneously, and the device-to-device communication feature that relays transmissions to devices in poor coverage are examples of synergies. In the future, these two features can be specified for NB-IoT using the same concepts and experience that were used to develop them for LTE MBB.

Conclusions

NB-IoT is the 3GPP radio-access technology designed to meet the connectivity requirements for massive MTC applications. In contrast to other MTC standards, NB-IoT enjoys all the benefits of licensed spectrum, the feature richness of EPC, and the overall ecosystem spread of 3GPP. At the same time, NB-IoT has been designed to meet the challenging TCO structure of the IoT market, in terms of device and RAN cost, which scales with transferred data volumes.

The specification for NB-IoT is part of 3GPP release 13 and it includes a number of design targets: device cost under USD 5 per module; a coverage area that is seven times greater than existing 3GPP technologies; device battery life that is longer than 10 years with sustained reachability; and meet a capacity density of 40 devices per household.

As NB-IoT can be deployed in GSM spectrum, within an LTE carrier, or in an LTE or WCDMA guard band, it provides excellent deployment flexibility related to spectrum allocation, which in turn facilitates migration. Operation in licensed spectrum ensures that capacity and coverage performance targets can be guaranteed for the lifetime of a device, in contrast to technologies that use unlicensed spectrum, which run the risk of uncontrolled interference emerging even years after deployment, potentially knocking out large populations of MTC devices.

The first standard development of 5G radio access is currently underway, with system deployment targeted for 2020. In this context, the ability to future-proof additional technologies like NB-IoT is a top priority. In the ongoing discussions in 3GPP surrounding 5G, LTE will continue to be an integral part of radio networks beyond 2020, and so, NB-IoT's resemblance to LTE safeguards the technology from diverging evolution paths.✪

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SECOND WAVE CONSOLIDATION ✱

THE CENTRAL OFFICE OF THE ICT ERA: AGILE, SMART, AND autonomous

Network architecture is undergoing a massive transformation, which in turn is having an impact on the role of the central office. Enabled primarily by virtualization and SDN technologies, network architectures are becoming more flexible, with improved programmability and a greater degree of automated behavior. In combination with technology enablers such as the increased reach offered by fiber, automation of provisioning and orchestration, and improvements in the performance of generic hardware, network transformation has provided operators with the opportunity to rationalize and consolidate infrastructure. The next generation central office will introduce intelligence and service agility into the network through disaggregation.

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THE CENTRAL OFFICES (COs) of fixed networks and the mobile switch offices (MSOs) of mobile operators house the networking functionality, management, and compute power needed to provide voice and data services to enterprise and residential subscribers. To route traffic efficiently, COs are distributed throughout the entire geographic region served by the network, and provide operators with a key asset: local proximity to their subscribers.

Traditionally, the location of a fixed-line CO has been determined by the reach constraints of the access technologies used in the last mile – from the CO to the subscriber (residential or enterprise). Until recently, copper was the predominant media, and so the location of the CO has been dictated by the maximum reach of the copper pairs supporting POTS equipment in the home or at the enterprise premises. Although copper is no longer the primary choice for access media (or even present in many cases), the location of COs still reflects the original distance constraints. As a result, even mid-sized cities with around a million subscribers are served by hundreds of COs, and it is still common for these to be placed in a grid-like manner, spaced a couple of kilometers apart. In rural areas with low population density, fixed-access technology reach is also the main factor for determining location, explaining why the ratio of subscribers to COs in rural areas tends to be low.

The first wave of CO consolidation and centralization came about during the digitization of POTS. Digitization resulted in a reduction in size or functionality of many city and rural COs, and in many places, they were replaced with small concentrators connected to a smaller number of more centralized COs.

The location of a CO is significant; when positioned in close proximity to users, certain services can be provided to local groups of subscribers in a highly efficient manner. This capability is one of the primary differentiating assets of the access operator.

Likewise for mobile networks, the optimal placement of an MSO takes location constraints into consideration, and the critical factor for mobile is access to the base transceiver station (BTS). Originally, cable was used as the primary media for BTS access, shifting in recent years to use of high-capacity fiber TDM circuits. So, for the same geographic area and subscriber base, MSOs tend to be more centralized compared with fixed COs.

Today, a medium-sized city can be served by just one or two MSOs, but possibly hundreds of COs. In rural areas, however, MSOs tend to be sparse or even nonexistent. Operators running converged fixed and mobile networks tend to house MSOs within existing COs, rarely opting for new builds in dedicated locations.

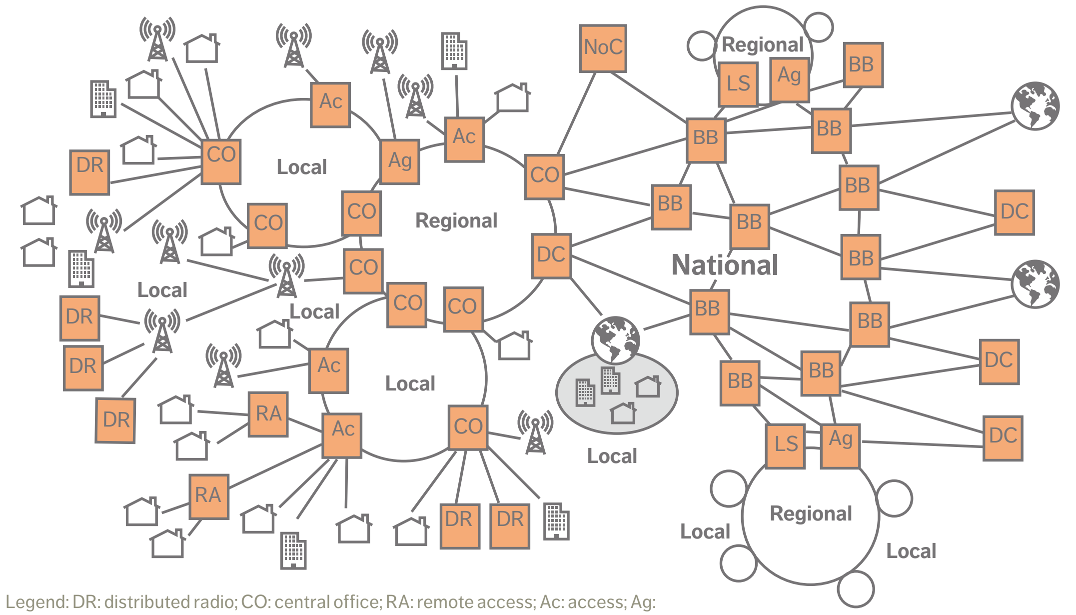
Figure 1 illustrates the local, regional, and nationally tiered structure of COs. Fixed COs have two or more progressively centralized tiers, which originally provided inter-office calling capability to avoid the need for a full CO mesh. Higher-tier COs have extensive transmission trunking from lower-tier and access COs, which is significant, as this architecture may be utilized for the placement of next generation central offices. MSOs can be colocated with a subset of COs or be deployed independently as local and regional COs. End sites

> THE LOCATION OF A CO IS SIGNIFICANT; WHEN POSITIONED IN CLOSE PROXIMITY TO USERS, CERTAIN SERVICES CAN BE PROVIDED TO LOCAL GROUPS OF SUBSCRIBERS IN A HIGHLY EFFICIENT MANNER

Terms and abbreviations

ACL — access control list | **API** — application programming interface | **ARPU** — average revenue per user | **BGP** — Border Gateway Protocol | **BNG** — Broadband Network Gateway | **BSC** — base station controller | **BSS** — business support systems | **BTS** — base transceiver station | **CDN** — content delivery network | **Clos** – Non-blocking, multistage switch fabric formalized by Charles Clos | **CMS** – cloud management system | **CO** — central office | **COTS** — commercial off-the-shelf | **CPU** — central processing unit | **DOCSIS** — Data Over Cable Service Interface Specification | **DSL** — digital subscriber line | **GPON** — gigabit passive optical network | **HLR** — home location register | **I/O** — input/output | **IGP** — Interior Gateway Protocol | **IoT** — Internet of Things | **ISP** — internet service provider | **M2M** — machine-to-machine | **MAC** — media/medium access control | **MME** — Mobility Management Entity | **MPLS** — multi-protocol label switching | **MSO** — mobile switch office | **NETCONF** — protocol to install, manipulate, and delete the configuration of network devices | **NFV** — Network Functions Virtualization | **NGCO** — next generation central office | **NIC** — network interface card | **NMS** — network management system | **NVGRE** — Network Virtualization using Generic Routing Encapsulation | **ODL** — OpenDaylight | **OLT** — Optical Line Termination | **ONIE** — Open Network Install Environment | **OSS** — operations support systems | **OTN** — optical transport network | **OTT** — over-the-top | **PON** — passive optical network | **POTS** — plain old telephone service | **P-GW** — packet data network gateway | **P/S-GW** — packet data network/serving gateway | **RNC** — radio network controller | **SDN** — software-defined networking | **SFP** — small form-factor pluggable | **S/GGSN** — serving/gateway GPRS support node | **vBNG** — virtual Broadband Network Gateway | **VIM** — virtual infrastructure manager | **VoD** — video on demand | **vSwitch** — virtual switch | **VXLAN** — Virtual Extensible LAN | **XaaS** — anything as a service | **XMPP** — Extensible Messaging and Presence Protocol

SECOND WAVE CONSOLIDATION ✱



Legend: DR: distributed radio; CO: central office; RA: remote access; Ac: access; Ag: aggregation; DC: data center BB: backbone; LS: local switching

Figure 1 CO tiers and distribution

connect to COs and MSOs through intermediate transport aggregation sites.

The COs house POTS and DSL/PON access equipment, and to a lesser extent, IP/Ethernet routing and switching capabilities for residential and enterprise services. MSOs house radio aggregation nodes, such as BSCs and RNCs, as well as transport switches. Some of the MSOs may house additional 3GPP core functions such as S/GGSN and P/S-GW, as well as control-plane functions such as MME and HLR serving multiple geographic areas. Other core functions can, however, be placed elsewhere, for example in purpose-built regional or national DCs.

In this second wave of CO consolidation, the fundamental internal structure and functionality provided at each site will change, and use of new technologies will either result in fewer sites or greater capacity. The term next generation central office (NGCO) has been adopted by the telecom industry to refer to the future central offices that will support both fixed and mobile operations. Compared with its current CO counterpart, the NGCO will be able to serve more subscribers, implement access functions in a more IT-centric way, and support and locally house new, flexible data services. The NGCO will function like a highly automated mini data center, requiring less space, power, and cooling than the set of traditional COs it replaces.

Why transform?

In addition to the constant need to reduce opex and capex, fixed and mobile operators continually face new challenges as technology and user demands change. Network transformation and changing subscriber traffic patterns have created new challenges in terms of the services operators offer, and perhaps more significantly, the services that operators would like to offer, and how to provide them in the shift toward the more attractive anything-as-a-service (XaaS) business model.

The shift from voice to data services and the corresponding massive increase in OTT traffic have put pressure on networks. Changes in user behavior, with preferences shifting to use of bandwidth-hungry data services, and video consumption require a revolutionary change in the way existing CO- and MSO-based network architectures are structured.

Traffic patterns and demands

The annual growth rate of traffic carried by mobile and fixed networks has risen massively over the past five years. In addition to increasing traffic volumes, meeting the ever more stringent demands placed on network performance characteristics, such as latency, is necessary to support emerging industry applications. Technology improvements made in fixed-network access and the mobile industry (as 5G systems evolve) will enable networks to cope with growing traffic volumes and performance demands. But, as network capabilities increase, user expectations and the demand for more capacity and bandwidth will also inevitably rise.

The increase in traffic volumes and performance demands can be predicted and planned for, but changing traffic patterns due to changing subscriber habits is complicating network architecture in a new way. As networks become more flexible, user-to-user and machine-to-machine flows will become more widespread, adding new dimensions to the traditional user-to-server traffic-flow pattern. Factor in the massive expansion of the Internet of Things (IoT) and the result will be an explosion in the number of flows and routes that networks will need to support.

With static or declining ARPU, the question facing many operators is how to invest in networks so they meet constantly rising performance demands.

Technology provides some useful steps that can help answer this question. For example, where possible and necessary to meet latency requirements or lower backhaul costs, self-served and partner content, such as video, and subscriber-associated IP service delivery points – P-GWs, BNGs, and multi-service edge routers – can be moved closer to the user. Traffic not served by the access operator can be offloaded to other ISPs, transit carriers, or OTT content providers that are closer to the access domain, rather than hauling it back to more centralized interconnection points. Similarly, instead of hubbing enterprise transport traffic through large centralized routing points, a more optimal way to route this type of traffic is through distributed routing points in the network. Shifting traffic around like this will dramatically alter the ratio of locally terminated traffic to transit traffic and requires the NGCO to provide support for routing and service functionality well beyond the capabilities of the traditional CO.

Efficient rollout of services

To take advantage of the revenue streams created by massive traffic volumes, tough performance targets and new traffic patterns, networks need to be able to support efficient rollout of services. Network flexibility is key here, enabling operators – and indirectly subscribers – to modify services to match their evolving needs, scale them easily, and be able to specify and change the location of service instantiation. Provisioning mechanisms need to be highly efficient, low opex and capex are essential, and, as time to market is crucial, high feature velocity is vital.

Access operators offer end services such as web applications, CDNs with their associated content caches, and bump-in-the-wire services including parental control filtering, as well as transport services such as enterprise connectivity or internet access, or a combination of both. More advanced services require support for service chaining that can be dynamically customized on a per-subscriber basis.

Public and private cloud-based XaaS is an attractive offering for both enterprise and non-enterprise customers, but requires support for multi-tenancy environments.

SECOND WAVE CONSOLIDATION ✱

“THE INCREASED PENETRATION OF FIBER IN THE LAST MILE IS PERHAPS THE MOST SIGNIFICANT FACTOR IN THE SHIFT TOWARD FEWER AND MORE CENTRALIZED NGCOS”

By appropriately locating these services in NGCOs, carrier networks will become part of a distributed and intelligent cloud resource, supplementing larger, centralized data centers.

Software development and deployment life cycle

A typical service life cycle starts with development and verification before moving on to wide-scale deployment in the network.

Efficient service life cycle depends on two key factors: short time to market and deployment flexibility. Time to market can be minimized through a homogeneous software environment that enables deployment on existing network infrastructure without the need for hardware modification. Deployment flexibility is needed to enable elastic capacity scaling, dynamic service chaining, and the deployment of services in new locations.

Key to implementing these factors in the NGCO is virtualization of the compute platform on which services run, so that the traditional coupling of software to specific hardware can be removed. Decoupling provides a homogeneous development and deployment environment that is suited to an automated life cycle.

Technological enablers

Fiber reach

The increased penetration of fiber in the last mile is perhaps the most significant factor in the shift toward fewer and more centralized NGCOs. Connectivity over the last mile may be delivered by a PON. This might come in the form of fiber, or as a hybrid solution in which a relatively short copper extension using VDSL or DOCSIS technology extends the fiber from the NGCO to the curb.

As an enabler, fiber applies primarily to the central offices for fixed services, as mobile offices already tend to be positioned to operate with long-reach access technologies.

Virtualization

As it decouples applications from the underlying hardware platform, virtualization is one of the key enablers for flexible service and function deployment.

With good orchestration, virtualization technologies enable most types of workloads to be consolidated on common multi-core compute platforms. Further reduction of hardware in the NGCO can be achieved by pooling workloads on a common compute resource, and additional power savings can be gained through dynamic workload reassignment.

The significance of virtualization in future carrier networks is clearly reflected by the massive effort being put into this area by operators, vendors, and standardization bodies. The heightened focus on all aspects of virtualization bodes well for the acceleration of its adoption.

Automating the VNF life cycle

Automated orchestration of virtual functions' instantiation, capacity elasticity, and function termination are critical network capabilities that enable functions to be deployed quickly and flexibly in multiple, geographically distributed NGCOs.

Orchestration is central to the operation of any virtualization environment offering multi-tenancy – whether it is for an operator's many internal tenants, or external residential and enterprise tenants.

Compute performance

The continuous improvements in compute performance can be attributed to a number of different technologies. Cores, for example, have become faster, the core per socket ratio has risen, on-chip caches have become both larger and faster, and access times to peripheral memory and storage have dropped dramatically. Today, it is fairly common for an individual CPU to contain tens of cores, each running at 3GHz on COTS hardware, with single, dual or quad sockets. In addition, I/O speeds have increased, enabling modern servers to support dual (and possibly more) 40Gbps NICs.

The increases in compute and I/O performance have in turn widened the set of functions that might benefit from virtualization. And so, network design is no longer restricted to the virtualization of traditional IT and control-plane intensive workloads, but can be expanded to include traditional telecom network functions that demand high user-plane performance, such as virtual routers and virtual subscriber gateways including virtual BNGS and P/S-GWS.

As compute capabilities continue to improve, an equivalent reduction in the hardware footprint of access functions will occur. This not only brings benefits in terms of cost and environmental impact, but also enables functions that benefit from proximity to the user, previously deployed in more spacious DCs, to be distributed and deployed in the NGCO.

DC switching fabric

To virtualize network functions and other workloads as far as possible, the NGCO obviously needs appropriate compute and storage capacity. Emerging DC fabrics – based on merchant silicon leaf-and-spine switches – that are scalable, and offer high capacity at low cost, provide just the right kind of internal network design between compute- and-storage components and the physical WAN and access gateways.

Most NGCO fabrics will be configured as non-blocking clos [1] networks, possibly with under- subscribed dimensioning, even though such a structure is not strictly required.

Software-defined networking

Applying the concepts of SDN to a network makes it centralized, dynamically provisioned, and programmable. The agility and flexibility SDN offers will be critical in providing new and multiple-service operators with the capability to offer whatever services they like to their subscribers.

Key architectural components

Figure 2 shows the location of the NGCO and how it is connected to the fixed and mobile services it offers to subscribers through the various access domains. The diagram also includes an abstract representation of the internal structure of the NGCO and its connections deeper into the network. The orchestration component manages the functions and infrastructure of the internal office as well as certain external entities such as access routers.

Infrastructure

The NGCO infrastructure consists of three major components:

- the switching fabric that links all other components together
- gateways – to the access domain and the WAN
- servers and storage

Initially, non-virtualized bare metal appliances that perform specific functions will also be part of the infrastructure. These appliances might be incorporated into the gateways or be implemented on separate hardware platforms, depending on the capacity of the gateway and how well the hardware performs.

Switching fabric

The structure of an NGCO may use overlay/underlay design principles or adopt a more traditional approach. In an overlay/underlay design, the switching fabric forms the underlay and is agnostic of service endpoints. In traditional architectures, the switching fabric is fully aware of the service endpoints. The size and scale of the fabric varies according to the requirements and location of the office. For example, a small NGCO serving tens of thousands of users may consist of just a few switches

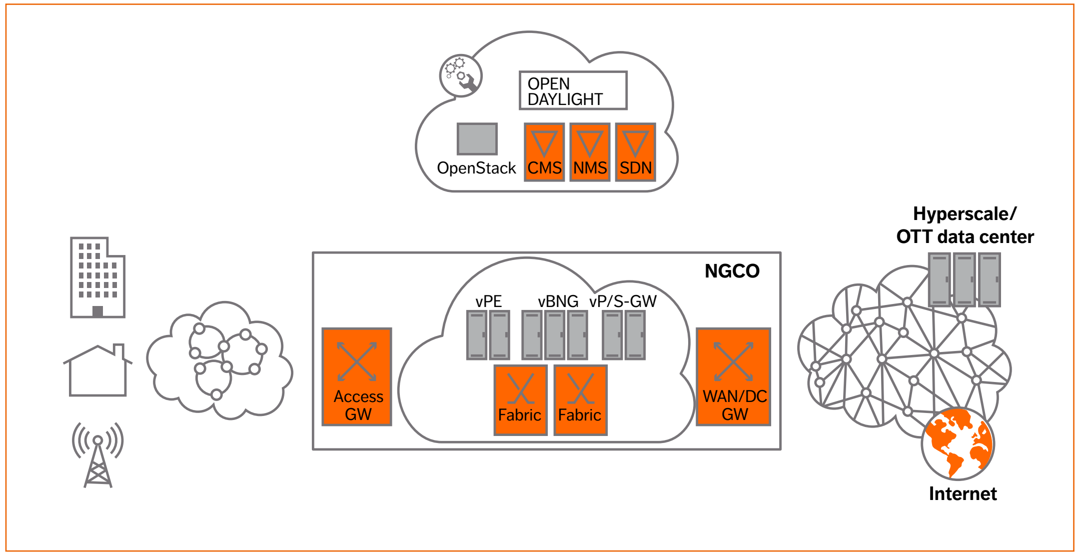


Figure 2 The NGCO in the operator's network

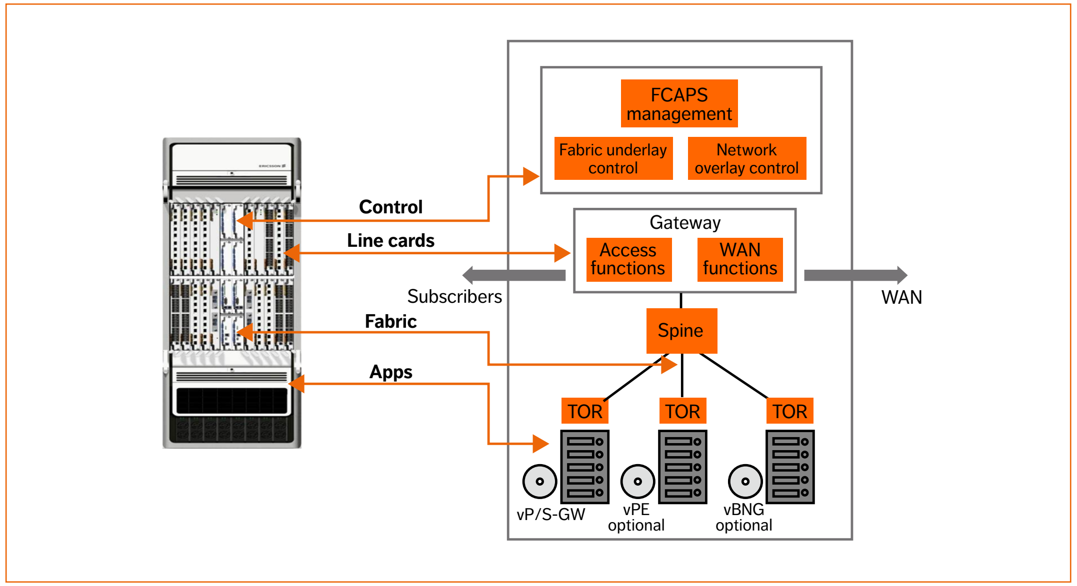


Figure 3 Disaggregation of routing functions

and support a minimum set of local functions, whereas larger offices may include a switching fabric capable of supporting extensive local services for millions of subscribers.

The structure of the fabric, especially when it comes to larger offices, is likely to be based on common data-center design practices, with an underlay clos architecture, using a cluster of leaf-and-spine switches with same-length links, offering potentially deterministic delay and latency. In a clos underlay, load balancing within the fabric is achieved by utilizing the multiple paths between source and destination. Either centralized SDN controllers or distributed routing protocols such as BGP or IGP will be used to build the forwarding, routing, and switching tables. To build the fabric underlay for large-scale NGCOs, the industry preference is leaning toward the use of distributed routing protocols, as they are simplistic and have a proven track record.

Merchant silicon-based white boxes can be used for fabric switches, especially when providing a simple underlay. These boxes tend to be less capable but often have a lower price-to-bandwidth ratio than traditional switches. White boxes offer entirely decoupled networking OS and hardware, and by using a tool such as the Open Network Install Environment (ONIE), for example, the installed network OS can be easily swapped out with another one – allowing operators to load the OS of their choice onto installed hardware. So, white boxes not only contribute to reducing costs; they perhaps more significantly provide network programmability and flexibility.

Shown in *Figure 3*, the NGCO fabric conceptually represents a disaggregated router that can be readily scaled out by adding leaf-and-spine switches as needed. The fabric may need to support a number of underlay technologies including IP and Ethernet, and MPLS may be required, especially in carrier domains, to ensure operational simplicity and seamless end-to-end interoperability with the installed base.

In the event of a switch failure, the fabric automatically reroutes traffic through the remaining switches until the failed switch has been manually replaced and auto-configured by a fabric manager, allowing the system to operate without having to wait for a maintenance window.

Optimum traffic management requires a holistic and real-time view of the available network bandwidth and traffic patterns. Flow statistics are collected at regular intervals, and when analyzed, provide the information needed to detect and avoid congestion, guarantee better utilization of fabric resources, and administer prioritization policies.

Gateways

Access and WAN gateways act as infrastructure gateways, and tend to be connected to special leaf nodes. The WAN gateway function could alternatively be implemented using spine switches.

Access gateways that terminate customer access links may require extended capabilities such as deep buffers, traffic management and other more advanced QoS capabilities, large forwarding tables, and ACLs that are not usually present in merchant silicon-based white boxes. Access gateways terminate different access technologies such as DOCSIS and GPON OLT. OLT functions can be virtualized with the MAC layer and the optics separated from the control- and management-plane software. The hardware part of the gateway can be implemented on a small SFP form factor, while the software part can be virtualized and hosted on any server within the CO.

Using a variety of communication protocols (such as IP, MPLS, and OTN/WDM), WAN gateways connect central offices with other NGCOs and COs, central and regional data centers, as well as other carriers and the wider internet.

Compute and storage

The geographic closeness of the NGCO to users provides a strong incentive to house certain functions and services that benefit from this proximity in the NGCO. Compute and storage resources exist in the NGCO to run virtualized network functions such as vBNG and vP/S-GW, as well as more traditional services such as VoD, with local caching.

The general purpose nature of compute resources deployed in the NGCO is key, as any network function or service can be instantiated on them, supporting

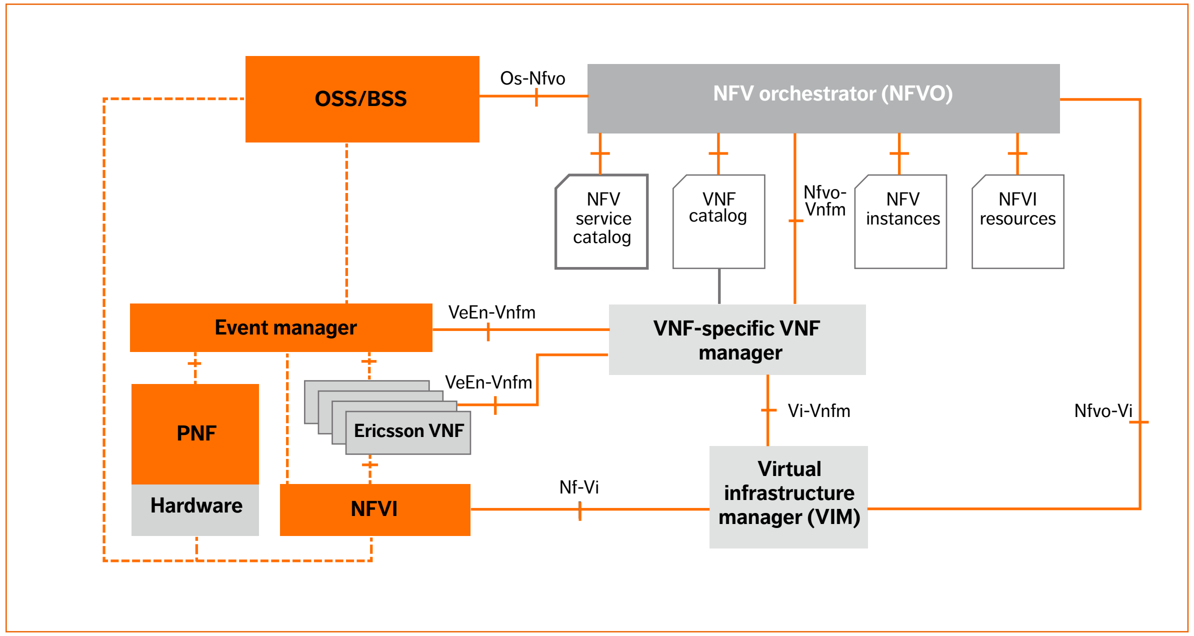


Figure 4 ETSI NFV reference architectural framework

the break away from traditional hardware and software coupling.

The amount of compute and storage located in a given NGCO will depend on its size and operator preferences for centralization versus decentralized function deployment. Offering cloud services, for example, requires additional compute and storage, which in turn increases the size of the NGCO.

Overlay services

If the NGCO implements overlay services using an underlay switching fabric, an overlay encapsulation technique is required. This technology can also be used to provide tenant isolation to operator-internal stakeholders and subscriber isolation for NGCOs with cloud services.

Common encapsulation technologies include VXLAN, NVGRE, and MPLS VPNs, and can be implemented virtually in vSwitches or in hardware on leaf and possibly gateway nodes if higher performance is required.

Regardless of the location and type of overlay technology used, configuration will be automated by an overlay controller coupled through northbound APIs to the automatic provisioning of any tenant-related functions. For example, the overlay controller could be ODL-based coupled to OpenStack through Neutron APIs. The same APIs can be used by additional applications such as the OSS/BSS. The overlay controller communicates with virtual network switches or bare metal devices (gateways and leaf switches, for example) preferably through

open southbound interfaces such as OpenFlow, XMPP, or NETCONF.

Virtualized network functions

In today's COs, traditional network functions and workloads, such as caches and webservers, run on vertically integrated platforms. In the NGCO, these elements will be run as virtualized network functions on COTS hardware.

NFV technology makes it easier to create and scale separate logical nodes and functions, and if necessary, these elements can be isolated for use by a specific tenant. This is the concept of network slicing. Network slices are individually designed to meet a specific set of performance requirements tailored to the application running on the slice. The virtual infrastructure of a slice is isolated from other slices to ensure that all slices of the network run efficiently and performance targets are met. The NFV approach provides the flexibility needed to provision network resources on demand, and to tailor slices to specific use cases, enabling operators to deliver networking as a service. The beauty of network slices lies in their ability to be optimized to suit the application. In other words, high-availability services can run on slices optimized for resilience to hardware and software failures, whereas an M2M signaling-intensive application, for example, can run on a low-latency, low-bandwidth slice.

Automation

In the NGCO, all key operational components are automated. This removes the need for manual configuration, which is prone to error, costly, and time-consuming.

The fabric manager oversees the automated parts of the NGCO, configuring and managing the underlying fabric switches, and supervising the performance of the fabric. The fabric manager continually and automatically monitors the physical fabric node-and-link topology, it validates the physical cabling, and configures leaf-and-spine switches with associated protocols and policies. The fabric manager may use DevOps tools such as Chef or Puppet for initial configuration and software management tasks

> “THE GEOGRAPHIC CLOSENESS OF THE NGCO TO USERS PROVIDES A STRONG INCENTIVE TO HOUSE CERTAIN FUNCTIONS AND SERVICES THAT BENEFIT FROM THIS PROXIMITY IN THE NGCO”

(LLDP configuration, management addressing, and OS component upgrades), after which programmatic interfaces such as NETCONF/YANG can be used to configure network protocols, QoS policies, and statistics on the interfaces. For centralized SDN-based cases, the fabric manager can use OpenFlow to configure the necessary forwarding entries in the underlay switches.

Service orchestration

Service orchestration automatically instantiates applications and configures network services according to service-level specifications. Automation of these tasks can dramatically reduce the time to instantiate or add new devices or services to the network, which increases network agility, making real-time service provisioning possible.

Migration

For most operators, the migration of network architecture from the current CO deployment to one based on fewer NGCOs will be gradual. While some NGCOs will be built as greenfield deployments, for the most part, existing COs will evolve, requiring the coexistence of decoupled SDN/NFV equipment, together with traditional, tightly coupled hardware and software. During the migration/coexistence period, management and orchestration components need to be able to support the heterogeneous (coupled/decoupled) environment; by, for example, abstracting the differences between the two architectures, and using common northbound interfaces to other systems, such as end-to-end service orchestration and OSS/BSS.

Throughout the period of coexistence, network functions will be physically and virtually instantiated with capacity and subscribers pooled across both, and as *Figure 4* shows, orchestration systems will be required to support both traditional and decoupled architectures.

Conclusions

Network architecture is undergoing a massive transformation in terms of increased levels of automation and programmability. This transformation has been enabled by a number of technologies, but primarily by the disaggregation of software and hardware. The transformation is being driven by new business opportunities, expected gains in operational efficiency, and the need for rapid time to market for services. As the underlying technologies – virtualization and SDN – become more mature, the rate of transformation will rise.

The next generation central office, or NGCO, has been designed to take advantage of the gains brought about by a decoupled network architecture. The benefits for operators come in the form of network intelligence, flexibility, and ease of scalability, all of which bring opex and capex benefits.

The NGCO is basically a mini data center that provides converged fixed and mobile services. Compared with a traditional CO, the NGCO can serve a larger subscriber base across a wider geographic area. The NGCO has been brought about through:

- reduced CO density, as a result of greater distances achievable by fixed access technologies
- the introduction of SDN/NFV technologies
- advancements in hardware technologies in terms of low-cost, high-throughput switches
- infrastructure automation and service orchestration

Architecturally, deploying the NGCO as a mini data center introduces a greater level of intelligence into the network in a distributed fashion, as applications are replicated, or shifted, from centralized data centers out to NGCOs. Compute resources in the NGCO can be used for running applications such as rich media and rendering, or latency-sensitive gaming apps. With these capabilities, the NGCO will become part of a distributed, intelligent cloud resource.

The NGCO brings with it a number of savings, requiring less space, power, and cooling than the sum of the individual traditional COs they replace. On-site staffing requirements should be reduced, as provisioning and many aspects of maintenance are controlled remotely and automated. Overall, the NGCO will result in fewer central offices or increased access coverage and service consolidation, with reduced need for new real estate as equipment continues to compact. ✪

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Five trends

shaping innovation in ICT

Keeping up with the relentless pace of change in the ICT industry is a daily challenge for modern tech companies. The key to long-term success lies in the ability to understand change almost before it occurs and seize the opportunity to shape evolving technologies.

Tech companies often gain competitive advantage by causing market disruption through their ability to understand and act on technology trends. Like waves in the ocean, it's much easier to ride these trends if you can see them coming and read them right. (But of course, true technology leadership happens when you start making your own waves.)

As I see it, there are five key technology trends that will stimulate innovation within the ICT industry in the coming year, creating new value streams for consumers, industries and society. All five pivot around a technology-enabled business ecosystem made possible through a universal, horizontal and multipurpose communications platform.

BY ULF EWALDSSON, CTO

#1

SPREADING INTELLIGENCE THROUGHOUT THE CLOUD

Distributed machine intelligence moves into the cloud

#2

SELF-MANAGING DEVICES

Intuition, self-learning, and increasingly autonomous devices

#3

COMMUNICATION BEYOND SIGHT AND SOUND

Human interaction augmented by tactile internet

#4

FUNDAMENTAL TECHNOLOGIES RESHAPING WHAT NETWORKS CAN DO

New materials and manufacturing techniques enhance networking capabilities

#5

WEAVING SECURITY AND PRIVACY INTO THE IoT FABRIC

Automation makes security controls real-time and proactive

#1

Spreading intelligence throughout the cloud

CONNECTED smart machines, such as robots and autonomous vehicles, are fundamental to the evolving Networked Society. Enhanced cloud architecture that can distribute and share machine intelligence will enable smart connected machines to work on an increasingly higher level.

■ Supported by advancements in artificial intelligence (AI) – particularly in the areas of big data analytics, machine learning and knowledge management – rapid progress has been made in terms of what smart machines can do. Developments in connectivity and cloud technologies are making it possible to distribute and share machine intelligence more easily, at a lower cost, and on a much wider scale than before.

When connected to the cloud, smart machines will be able to use the powerful computational, storage and communication resources of state-of-the-art data centers. Today's intelligent software robotics systems are capable of supporting repetitive administrative tasks with current development pushing toward advisory tasks. Cloudification shifts the capabilities of these systems into a new sphere that includes complex problem-solving and decision-making on a mass-market scale.

Connect, store, compute... and share

Shifting systems into the cloud enables communities of collaborating robots, machines, sensors and humans to process and share information. Each new insight collected within a community can be shared instantly, which increases the effectiveness of collaborative tasks, and improves performance throughout the system, with a common awareness of system state shared by all participants, as well as a shared knowledge base.

A distributed machine intelligence architecture offers lower implementation costs. Sharing a backbone of almost unlimited computational power makes it possible to build lightweight, low-cost robots and smart machines that require a low level of control and a minimum amount of sensors and actuators. Application-specific requirements related to responsiveness and speed will determine whether a local or global cloud is most suitable, and how much intelligence can be distributed.

Smart and mobile capabilities virtually everywhere

Intelligent clouds will create new value chains in many industry segments, but some of the forerunners include mining, agriculture, forestry and health care. New opportunities will open up for all organizations and people involved in the supply chain from the manufacturer to the customer. Consider an automated agriculture application. The application remotely controls farm machines to carry out various farming tasks. To harvest mature crops, for example, the system will control the necessary machines to cut, gather and transport them. Each individual machine will take local decisions to ensure secure completion of its set tasks, working in conjunction with all the machines involved in the harvesting. Weather

reports gathered from another distributed cloud application are used by the system to carry out harvesting in an optimal way. Contact with the farmer occurs only when participating machines cannot resolve issues themselves.

The harvesting example highlights just one of the many coming applications that will rely on multiple information sources, cloud, and distributed machine intelligence. To ensure scalability and widespread uptake of such applications, the challenge lies in the rapid development and proliferation of universally accessible mobile capabilities. 5G will provide a resilient, high-availability, low-latency network that offers applications with integrated computing and storage resources that are ideally placed to meet latency requirements. 5G is well matched to industrial robotics applications because, like other radio technologies, it removes the need for cabling and minimizes infrastructure adaptions, but it also offers identity management, optimum placement of resources, and encryption for security and privacy. ✪

#2

Self-managing devices

THE CONNECTIVITY ALLOWS OBJECTS TO BE SENSED AND ACTUATED REMOTELY, CREATING A BRIDGE BETWEEN THE PHYSICAL AND DIGITAL WORLD

COMBINING SENSORY DATA with AI techniques enables the data from massive numbers of sensors to be merged and processed to create a higher-level view of a system.

Connected smart devices will change our lives in many ways. These range from simple services that open your garage door as your car approaches, for example, to radically new business opportunities involving services yet to be invented and markets yet to be discovered. Combined with intelligent handling of data, smart devices can boost the productivity and profitability of any business. But to enable the deployment of billions of smart devices, the cost of managing and monitoring them needs to be low. Evolving software and communications technology are shifting toward the creation of autonomous and self-managing devices.

The Internet of Things (IoT) means automation and intelligence in everything that is connected. This implies that a collective intuitive behavior among a wide range of devices for a wide range of applications is possible in the future. The connectivity allows objects to be sensed and actuated remotely, creating a bridge between the physical and digital world.

It's the combination that triggers the effect

Beyond the physical devices embedded with processors, software, sensors, actuators, and connectivity, it is the combination of sensory data and AI that enables more effective and accurate interactions. It is by merging data from a multitude of sensors that a superior baseline for intelligent processing is created. These are the common denominators that push IoT development further.

From a connectivity perspective, two distinct and different use cases emerge. One extreme is the massive machine-type communication (massive MTC) that can support millions of connected devices such as energy meters and logistics tracking. Here, we are looking at device battery lifetimes beyond 10 years and cost reduction in the order of 80 percent as well as 20dB better coverage compared with present state-of-the-art solutions.

The other extreme is the critical machine-type communication (critical MTC), which entails real-time control and automation of dynamic processes in various fields such as vehicle-to-vehicle, vehicle-to-infrastructure, high-speed motion, and process control. Critical parameters to enable the performance required are network latency below milliseconds, ultra-high "five nines" (99.999 percent) reliability. The future network architecture needs to cater for both MTC scenarios.

Key technology advancements

The 2016 Ericsson Mobility Report *(https://www.ericsson.com/res/docs/2016/ericsson-mobility-report-2016.pdf)* predicts that there will be 28 billion connected devices by 2021. On the device side, the key technology driver is the evolution of sensors, actuators, processors, memories, and batteries. Beyond conventional electronics, we will see implementations of nanoscale technologies based on thin-film, graphene, and quantum sensors. We can expect any size and shape of device in the future.

Another emerging key technology is that of an advanced software toolbox leveraging advanced analytics, machine learning, and knowledge management with processing capabilities of real-time streaming data. Intelligent control logic is another interesting area. There is an increasing need for standardized platforms and software protocols. These will inevitably drive market consolidation, with massive cost savings and productivity gains as a result.

Effective connectivity and identity management are fundamental to the future network. These imply automated deployments, aggregated subscription management as well as embedded provisioning and control through the whole life span of the device.

What does this mean for the future role of networks?

IoT devices enable us to monitor sensors and automate a lot of processes. The added intelligence needed is a feature that will mainly be embedded in the network itself.

For IoT technology to live up to its promise and be applied on a massive scale throughout society, it must be built on a secure, global, telecom-grade network that is based on common standards. This will also ensure a healthy competitive and innovative ecosystem.

In terms of 5G, such an underlying network infrastructure is already in place – ready to show how well it is scaling and how its cost-efficiency properties support IoT applications. 5G offers both super-high bandwidth with ultra-low latency and extreme battery life for devices. By combining cloud intelligence with a powerful but energy-efficient wireless connection, even very simple and inexpensive devices can be made smart and generate great business value.

TECHNOLOGY TRENDS

#3 Communication beyond sight and sound

COMMUNICATION will evolve in a highly remarkable way over the coming years, as interaction between human beings and machines evolves to include additional experiences and senses. The internet you can feel is on the horizon.

■ Today, 2D video is the most advanced form of communication people use to connect with each other. In the future, people will be able to participate in distant business meetings or attend a family gathering by sending an augmented 3D selfie. I am sure many people are looking forward to the day it will be possible to attend events such as Mobile World Congress, the FIFA World Cup, or the Super Bowl virtually.

Emerging technologies in the fields of the tactile internet, virtual reality and augmented reality – supported by 5G network evolution – are showing signs that the ability to experience an event virtually is no longer science fiction, but a feasible reality, and indicate a giant step forward in innovation.

The tactile internet is founded on the visionary principle that all of our human senses can be embedded in human-machine interaction. Using haptics (interaction involving touch), remote experiences can be a near real-time representation of reality. To accomplish such realistic remote experiences, however, the loop connecting the disciplines of robotics, AI, and communications needs to be closed and near-zero latency requirements will need to be met.

Virtual and augmented reality (VR and AR) are expected to become integral technologies of the Networked Society, potentially disrupting the consumer electronics market.

Pushing the boundaries of traditional physics

To close the robotics, AI, and communications loop quickly, Ericsson has started a collaboration on the tactile internet with King's College London. As the research team puts it, "We need to beat the limits of the traditional laws of physics, as even the speed of light is not fast enough to enable these kinds of applications."

In this context, tactile communication enables haptic interaction between control and machine with visual feedback. Technical systems will need to support audiovisual interaction, and enable remote robotic systems to be controlled with an unnoticeable time lag. End-to-end, components other than the physical distance separating control from machine add to the total system delay. For instance, video coding and rendering require a substantial amount of computational power, and so these components increase overall system delay.

This type of next-generation communication will contribute to the resolution of complex challenges that arise in many sectors such as education, health care, personal safety, smart city, traffic management and energy consumption. Some business-related examples include virtual stores, interactive 3D design labs, training, interactive entertainment, and enterprise communication. Presently, the gaming industry is the primary incubator for AR and VR.

Not just raw speed – some intelligence too

Human-to-human and human-to-machine communications will put high demands on future networks. Solutions supporting high capacity and extremely low latency in combination with high availability, reliability, and security will define the characteristics of the network. In massive video distribution, for example, the need for capacity is created by certain application needs for high resolution, high dynamic range, and high frame rate, which in turn necessitate link speeds in gigabits per second. But it's not just about raw speed. Our research in this area has, for instance, investigated the idea of dividing the amount of transmitted data into priority hierarchies with different time requirements, transmitting only data that has been modified and anticipating changes.

#4

Fundamental technologies reshaping what networks can do

THE LAWS OF PHYSICS are the only real restriction on the development of communication networks. Ericsson is firmly committed to pursuing innovations that challenge present system limitations to help us reach beyond what is possible today.

■ While becoming increasingly versatile, the network's fundamental building blocks are also becoming much smaller, mimicking the way living things have evolved. The network of the future will be akin to the digital embodiment of an intuitive organism that is able to handle vast amounts of consciously intelligent automated resources. New materials in combination with innovative manufacturing technologies promise to radically enhance network capabilities.

Which technologies have the greatest potential to spur network evolution in the near future?

In the semiconductor area, a wide range of new materials and manufacturing technologies will soon become mainstream. New packaging and integration technologies offer substantially increased bandwidth in addition to power reductions.

The semiconductor industry is also at the cusp of leveraging new memory technologies that will be able to take on different roles in the system memory hierarchy, as well as offering substantial improvements in system input and output performance.

The semiconductor industry advances through continuous scaling of traditional CMOS. Major players are working on the 10nm node, and industry roadmaps include 7nm and 5nm manufacturing technologies. Advanced 2.5D/3D integration techniques for non-monolithic integration have the capability to offer a whole system function integrated on a single chip. These solutions are both cost and energy efficient. The introduction of multicore central processing unit solutions at power consumption equal to or lower than their predecessors is a predominant trend. Other trends include the development of various types of architectures aimed at significantly accelerating processing speed, such as massive parallel computing.

Electrons and light blending in new ways

Advances in silicon photonics allow for optical integration directly into the processing unit and other vital components in a communication network. Photonics will add properties such as low propagation loss, high data-transfer density, and excellent signal integrity. Bridging the gap between optical and electronic components, silicon photonics will shrink everything including the footprint, power consumption, and cost of high-speed network applications. Furthermore, silicon photonics will allow for greater disaggregation of functions, which opens up for more efficient hardware architectures, while enabling more aggregated data traffic.

“NEW MATERIALS IN COMBINATION WITH INNOVATIVE MANUFACTURING TECHNOLOGIES PROMISE TO RADICALLY ENHANCE NETWORK CAPABILITIES”

Qubits – small but powerful

Slightly further into the future, quantum computing promises to bring about an exponential increase in computational power. Quantum computing is a technology that builds on the quantum properties of elementary particles (qubits). Qubits can be entangled with each other and can take on intermediate values compared with ordinary bits, which can only be either 1 or 0. This way, a quantum computer can increase parallelism and radically reduce the computing efforts needed to address certain types of problems. Researchers have already succeeded in creating qubits within a semiconductor, and the first fully operational quantum computer was displayed at the end of 2015. One of the main challenges is to keep the quantum state unperturbed, which requires extremely low temperatures and very good insulation from the surrounding environment.

By matching the exponential expansion of the digital universe with computational power that also grows exponentially, we are confident that we will be able to continue to stay on top of future demands for communication. ✱

#5

Weaving security and privacy into the IoT fabric

IN A WORLD where everyone's personal and financial information is available online, cyber security and privacy are very serious issues for consumers, corporations and governments alike. And the rapid rise of wearables, smart meters, and connected homes and vehicles makes security and privacy more vital than ever.

■ The complexity and heterogeneous nature of future networks and connected devices will require security and privacy controls to be made an intrinsic part of every device, network, cloud and application. However, controls are only valuable if they can be managed in a fast and coordinated manner across all layers – preferably in an automated fashion, steered by policies and analytical insights rather than by the choices of an individual. Automated security and privacy management that is pervasive yet observable and auditable are the core characteristics that can enable the future Networked Society.

Weaving intelligence on three levels

Three layers of technology make it possible to weave security and privacy protection into every layer of ICT: actual security controls, security analytics, and an adaptive security posture.

Over the next decade, key security controls will include data sovereignty and novel identity management controls that are tailored to people and devices, as well as encryption technologies. Some encryption technologies are in the early phases of development but will begin to appear on the market in the next three to five years, as the underlying technologies mature. New root-of-trust technologies that are applicable to both physical and virtual environments also show great promise, and significant effort will be put into making them a reality.

Novel security analytics technologies can now provide insights that make it possible to create predictive security systems as opposed to reactive ones. These technologies could be used to create disruptive data management solutions in the near future, but for this to happen, we need to have context-aware security feeds and security analytics algorithms that correlate these feeds, often across multiple domains.

The third technology layer, the adaptive security posture, is achieved through automation, based on security analytics insights and policy-based automated orchestration of security controls.

It will all be built on trusted networks

No single industry player will be able to address all of these challenges on its own. Industry-wide collaboration, joint development, and standardization – including vendors, service providers, and users – will be essential in order to realize the vision of a secure Networked Society that protects business assets and everyone's privacy.

Traditionally, network service providers rank among the most trusted industry players. With this in mind, I believe that network service providers and their networks will be the foundation upon which the trust for everything else – devices, clouds, communications and applications – is built. At Ericsson, our focus is on enabling networks to play this key role across multiple industries.

PAVING THE WAY TO telco-grade PaaS

Independent of business, ways of working, or even technology adoption, the pressure on modern industries to shorten time to market through rapid development cycles is constant. The concepts of platform as a service (PaaS) and microservices – which have been gaining traction in the IT world – are deeply rooted in this need to cut development times. And the benefits are equally important in the telco domain. But there are gaps that need to be closed before PaaS is suitable for telco. Most of the challenges relate to the need for additional features that telco applications typically require. Once PaaS is telco approved, new applications will need to follow a number of design patterns, so that the full advantages of the platform-as-a-service approach can be realized.

EDVARD DRAKE
IBTISSAM EL KHAYAT
RAPHAËL QUINET
EINAR WENNMYR
JACKY WU

PAAS is a cloud service model that allows developers to build, run, and manage applications in a way that best suits their business needs, and most significantly, in a way that is independent of the underlying hardware or software infrastructure. Typically, PaaS enables developers to deploy code on top of a software stack that includes a runtime environment for one or several programming languages, an operating system, and basic services to build upon. PaaS provides the building blocks for automated testing, continuous deployment, as well as supporting the DevOps approach, and as such simplifies the development process and reduces time to market.

In the stack of cloud service models, shown in *Figure 1*, PaaS fits in between software as a service (SaaS) (which targets users with licensed software offerings) and infrastructure as a service (IaaS) (which addresses the management and sharing of hardware resources).

PaaS works with various cloud models: public, private, or hybrid. The hybrid model can, for example, be used by enterprises and telecom service providers to optimally combine the different handling needs of sensitive and non-sensitive workloads, where the common management interface enables some to be deployed on a private cloud and others on a public cloud – as shown in *Figure 2*. Latency-sensitive workloads, for example, or tasks that require security or control for proprietary data can be deployed on premises in a private cloud, while non-sensitive workloads can be deployed in a public cloud, maximizing agility and optimizing costs.

Depending on the level of automation and integration provided, PaaS solutions can be further divided into two categories: structured and unstructured. Unstructured platforms leverage basic container technologies or public PaaS offerings and are usually managed or monitored with homegrown tools. Technology-centric companies tend to favor such unstructured platforms, as they facilitate development and maintenance of solutions customized to meet business needs.

Structured platforms, on the other hand, come with built-in features such as orchestration, monitoring, governance, load balancing, and high availability. These characteristics make structured platforms suitable for enterprises or telecom service providers, and are the reason behind Ericsson's focus on structured PaaS.

> “STRUCTURED PLATFORMS COME WITH BUILT-IN FEATURES SUCH AS ORCHESTRATION, MONITORING, GOVERNANCE, LOAD BALANCING, AND HIGH AVAILABILITY”

The benefits brought by PaaS

What benefits PaaS can offer vary from business to business and from one application to the next, depending on whether it has been specifically designed for PaaS or whether it simply runs in a PaaS environment. The PaaS approach is well suited to application developers and vendors, but it can also be of great value to other users such as system integrators and service operators.

Some of the concepts used in PaaS, such as multiple application instances and component-based architecture, are established approaches in the telco domain. To keep the complexity of components at a manageable level, the telco domain has a long-standing tradition of modular design. However, designing applications specifically for PaaS increases the number of benefits for the different user groups.

Benefits for application developers

PaaS enables developers to focus on the business logic of their applications, as it frees them from the concerns associated with setting up the necessary foundation for deployment, testing, adaptation, and rollout. In doing so, PaaS enables innovation acceleration and rapid time to market.

Terms and abbreviations
IaaS – infrastructure as a service | **MMTEL** – multimedia telephony | **PaaS** – platform as a service | **SaaS** – software as a service | **SCTP** – Stream Control Transmission Protocol | **UDP** – User Datagram Protocol | **VNF** – Virtualized Network Function

Applications designed to run in a PaaS environment are likely to be less complex and consume less resources than their traditionally-programmed counterparts, as they do not need to re-implement the services that are provided by the platform. As a result, a PaaS application takes less time to start up than applications deployed on a full software stack. The simplified nature of PaaS applications brings benefits in terms of scalability, especially for those that are stateless.

Designing an application for PaaS with loosely-coupled internal and external interfaces makes it easier to manage life cycles for the components of an application and for the services they use in an independent manner. Deploying components that are loosely coupled not only simplifies an upgrade, it also reduces the complexity of validating an upgrade. Combined with the freedom to choose the programming language and runtime environment best suited to the task at hand, loose-coupling enables components to be replaced at any time with a different implementation – even in a different language – which in turn supports the gradual introduction of new technologies.

The PaaS framework provides common ways to expose and bind to services, which simplifies the deployment of new services. Service gateways and brokers can also expose external services, so they can be used by applications running inside or outside the PaaS environment.

The ease of integration of new services brought about by PaaS contributes to faster innovation, which is one of the model's primary benefits.

Benefits for system integrators

Some of the benefits that apply to developers also apply to system integrators. Loosely-coupled services and independent life cycles, for example, can simplify the testing and upgrade of components, as these tasks can be carried out separately. And the common binding and service exposure framework facilitates the integration of new services.

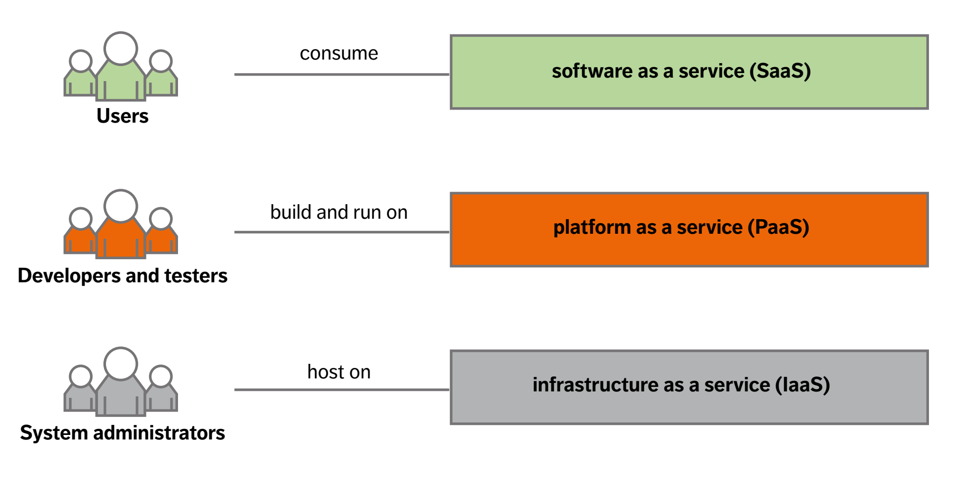

Figure 1
Cloud service models (from the point of view of the service consumer)

Benefits for service operators

A PaaS-designed application can scale quickly and easily with flexible workload deployment, which leads to optimal use of hardware resources. However, care should be taken when dealing with applications designed with large numbers of lightweight components that need to communicate with each other, to ensure that workload deployments do not negatively impact performance.

In general, security assurance and governance both benefit when applications run on a common framework that provides collective application management and supports intra-service communication. For example, the platform approach removes the need to manage masses of ad hoc security solutions and the rules governing how they apply to applications.

How do microservices contribute?

The software industry is currently experiencing a rise in the use of microservices and microservices architecture. And while PaaS and microservices are two separate concepts, viewing PaaS in combination with microservices and other concepts like containers and DevOps, can substantially increase the leverage of each of them.

Microservices is an architectural pattern and an approach to development. Essentially, this approach builds applications from (or deconstructs existing applications into) small parts – each with a single and well-defined purpose. To communicate, the parts (or microservices) use language- and technology-agnostic network protocols, and each part can be developed, maintained, deployed, executed, upgraded, and scaled independently. Technology choices are specific to the microservice and each microservice should be owned by a small team of developers to minimize the overhead of intra-team communication.

Overall, the ability to develop parts in an independent way enables rapid progress, allowing development to keep pace with market demands, and facilitates scaling of development.

Decoupling and independency between microservices is fundamental to a microservices architecture. Independence supports scaling over

> “THE ABILITY TO DEVELOP APPLICATION PARTS IN AN INDEPENDENT WAY ENABLES RAPID PROGRESS, ALLOWING DEVELOPMENT TO KEEP PACE WITH MARKET DEMANDS, AND FACILITATES SCALING”

multiple teams because it enables many small teams to work in parallel, with clear responsibilities, a large degree of freedom, and minimal interaction. Decoupling also enables the different parts of the system to evolve at their own pace.

Avoiding dependencies enables technology choices to be made on a per-microservice basis. As new technologies become available, they can be implemented appropriately without the need for a synchronized cross-microservice upgrade. As a result, each microservice can evolve at the right pace in a way that is most appropriate for a particular service: an efficient system that lends itself to the creation of ever-improving services.

While the advantages of a microservices architecture are apparent, in practice, this approach poses a number of significant challenges. To start with, the well-known fallacies of distributed computing [1] should be avoided. To perform a given task, a number of microservices are invoked sequentially, each of which contribute significantly to overall latency, making it more difficult to predict the the overall latency of a service. So, assuming, for example, that bandwidth is infinite, or that latency is zero can result in costly redesign work. Challenges include the overall complexity, both in development and in runtime, of a large, highly distributed systems. The ability to test a system is equally challenging, particularly when it comes to additional complex failure scenarios.

One way – and maybe the only way – to overcome the challenges surrounding latency is to accept that some parts of the system need to be designed with

> AS NEW TYPES OF IT APPLICATIONS EMERGE, SUCH AS IOT FOR VEHICLES, PLACING STRINGENT LATENCY DEMANDS ON COMMUNICATION LINKS, TELCO-LIKE REQUIREMENTS ARE BEGINNING TO APPEAR IN THE IT SPACE

miniservices, which are larger than microservices to allow for minimizing sequential communication, and consequently keep latency within required boundaries. When this approach is applied to appropriate parts of an application, the scope of what can be addressed with a microservices architecture can be broadened. And even if the application of miniservices can compromise some targets, the overall advantages gained tend to be fairly considerable.

Telco-grade and PaaS support

Existing PaaS environments have been designed to suit commodity IT applications, and as such, have not been built to meet the availability and latency requirements that are characteristic of telco applications. But as new types of IT applications emerge, such as IoT for vehicles, placing stringent latency demands on communication links, telco-like requirements are beginning to appear in the IT space. Enabling PaaS environments to meet telco requirements will benefit emerging IT applications, telco applications, as well as traditional IT applications – as long as the cost of providing common IT applications does not rise. In other words, adding a telco requirement, like five-nines availability, to the PaaS environment needs to be done in such a way that it doesn't impact performance or increase the setup complexity for other applications.

What then are typical telco requirements? Without constituting a formal definition, telco applications tend to be characterized by latency and availability (TL 9000 for example).

When it comes to latency, a requirement labelled soft real-time (outside of TL 9000) has been set for telco applications. This requirement is usually set to ensure that 95 percent of operations can be handled within a predefined latency budget – which for IMS is typically around 50ms per node in the traffic flow.

When it comes to availability, telco applications require five-nines availability. For PaaS environments, this level of availability implies that applications must be available 99.999 percent of the time, which corresponds to a maximum of five minutes of downtime in a year, including time for upgrades and other administrative operations.

When it comes to measuring availability, an application is deemed partly unavailable if it is out of service for more than 15s, with a loss of capacity in excess of 20 percent. If O&M functionality is lost, the loss is counted as 10 percent downtime for the whole node, even if the traffic part of the node is working.

To meet availability requirements, applications must be able to handle internal software and hardware faults without crashing. This is usually referred to as high availability support. A number of other factors also affect an application's availability, helping to determine whether or not it will attain the five-nines goal. Today's PaaS environments provide the necessary support in some cases, whereas development is still required in others.

To make PaaS telco grade, meeting availability requirements is not the only gap that needs to be closed; a number of others including logging support and security also need to be addressed:

Automation of management operations

System outages are often caused by human error during manual operation. Automating procedures minimizes this type of error, leading to improved availability. Many management operations in modern PaaS environments are already being automated, but further improvements are possible – such as single-command installation of an application that consists of a number of microservices.

Monitoring support

Through continuous monitoring of applications and the infrastructure, potential problems can be detected and rectified before they cause an outage. Today's PaaS environments include some basic monitoring support (typically checking whether or not an application is executing) but more advanced mechanisms are needed, especially for a microservice architecture, to handle scenarios where applications are in deadlock, or to cope with an internal crash, for example.

Logging support

When a fault does occur, the outage time until normal operations can be regained has a significant impact on overall availability. Access to information about the error is the key to rapid resolution. A presentation of the logs from a PaaS application – including services used – is needed to facilitate proper fault analysis. Existing PaaS solutions do not yet support such a feature.

Security

The use of role-based access to restrict actions on applications and the infrastructure reduces outages caused by a lack of the right kind of competence. This type of security measure also protects applications from malicious access, which reduces the risk of downtime. Current PaaS environments provide a degree of support in terms of access control, but additional backing is needed in the form of data encryption solutions, protocols that guarantee separation of traffic, and certificate handling, for example.

Tenant isolation

Strong support for tenant isolation covering all aspects of the infrastructure (compute, network, and storage) minimizes the impact a fault in one tenant application has on other tenants sharing the same physical resources. Current PaaS environments have sufficient support for tenant isolation.

Upgrade support

To ensure that capacity losses are kept below the 20 percent limit, application and infrastructure upgrades need to be carried out without affecting operation. For stateless applications, upgrades to individual instances of the application can take place without causing a disturbance. It must also be possible to upgrade the PaaS environment itself without affecting running applications.

Backup and restore

Saving the state of an application and corresponding software version enables an application to be restored quickly following a major failure, minimizing downtime.

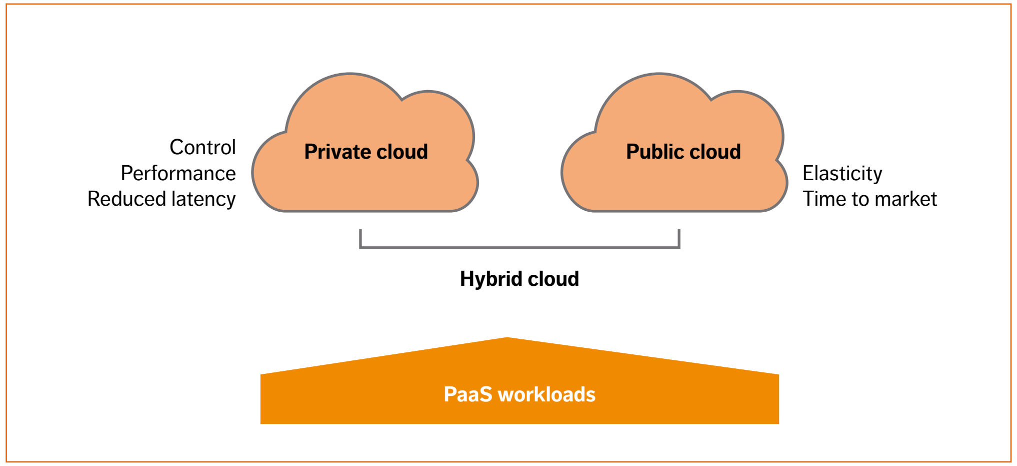

Figure 2
Deploying PaaS workloads in different types of clouds

Quick restart time
The ability to restart quickly reduces the outage time following a major problem. This capability is currently supported for stateless PaaS applications.

Independent restart
To maintain independence of infrastructure and application availability, restarting a PaaS environment in the event of an internal fault, upgrade, or other administrative process should not necessitate the restart of applications.

Network protocol support
Telco applications require support for a wider set of network protocols than typical IT applications. For example, UDP and SCTP are typical of the type of protocols that telco applications rely on, which are not found in traditional IT environments. Adding support for such protocols will expand the set of applications that can be deployed.

Alarms
Longer outages can be avoided by issuing an alarm indicating both that a problem has arisen and how to address it. This information needs to be collected in a consistent way from the application and the PaaS services being used.

Performance counters
A system that supports a rich set of performance counters preempts problems before they arise, and takes proactive measures to avoid outages. This information needs to be collected in a consistent way from the application and the PaaS services being used.

Trace support
If a fault occurs during operation, troubleshooting needs to be fast to minimize outage time. The trace output of applications is needed to facilitate proper troubleshooting. This information needs to be collected in a consistent way from the application and the PaaS services being used.

Soft real-time
The networking solution in the PaaS environment must be highly efficient with short round-trip times to fulfill the latency requirements of telco applications. Closing this gap is a significant improvement as doing so will increase the possibility of deploying a microservices architecture (one of the biggest concerns over such an architecture is the latency induced by multiple network hops).

Achieving telco on PaaS
This list of wanted features highlights the gaps that need to be closed to make PaaS suitable for all telco applications. But once these features have been added, what opportunities are likely to arise?

Building a new PaaS application
When writing a new application based on microservices architecture to run on PaaS, new design principles are needed to maximize the benefits. Some of these principles are borrowed from the twelve-factor app [2]. Among these principles, telco microservices need to:

- encapsulate a well-bound and well-defined function with a clear business need
- have separate life cycles – so they can be developed, delivered, installed, and upgraded independently
- be stateless – by storing data that needs to persist in a stateful backing service
- adhere to decoupled communication – using well-defined interfaces, network protocols, and asynchronous messages
- adhere to the share-nothing principle – data is not shared between the different instances of a microservice, enabling microservices to be scaled out by adding more processes
- be built for failure – to manage service failures or underperformance by supervising responses and throttling the requests, for instance.
- discover peers dynamically – by using the platform service discovery function to discover other microservices during runtime
- be technology agnostic – so that the choice of technology adopted for one microservice does not affect the choice of another
- fail fast: instead of implementing complex recovery mechanisms when severe errors occur, mini/microservice instances should fail with a clear error message stating the reason for the failure
- achieve high availability – by running in multiple instances, allowing one instance to take over if another one fails, and reading up state from the external store.

These principles should be complemented by state-of-the-art microservices design patterns [2].

Extending mature applications
The issue of handling legacy applications inevitably arises during discussions over the design of new applications. When evaluating the portability of mature applications to PaaS, a number of considerations need to be taken into account. Generally speaking, rewriting applications is not the best approach because it tends to be costly/time-consuming, it blocks the addition of new features, and opens the door for competitors to gain market share in the meantime. If the main reason for the adoption of PaaS is to gain development speed, redesign will not generate any market value if no new features need to be developed.

There are certain opportunities that PaaS adoption can take advantage of without the need for application redesign. If a mature application requires a new feature or if a market adaptation is called for, the feature could be implemented as a microservice on PaaS and accessed from the legacy application. This is indeed the recommended approach for shifting monolithic applications to a microservices architecture: to break out individual services one by one. When this approach can be taken, the best of both the IaaS and PaaS worlds can be captured: the performance of the existing application remains stable, while new features benefit from the rapid development PaaS offers, freeing up the design team to develop new features with additional business value – as illustrated in *Figure 3*.

Naturally, communication needs to be enabled between the main application and the new microservices. Some telco applications already have predefined interfaces that can be used to extend the node functionalities, such as the Parlay interface for the MMTEL application server. However, when no such interface exists, the main application and the microservice should be decoupled using an anticorruption layer. All patterns used for migrating from monolithic to microservices such as facades,

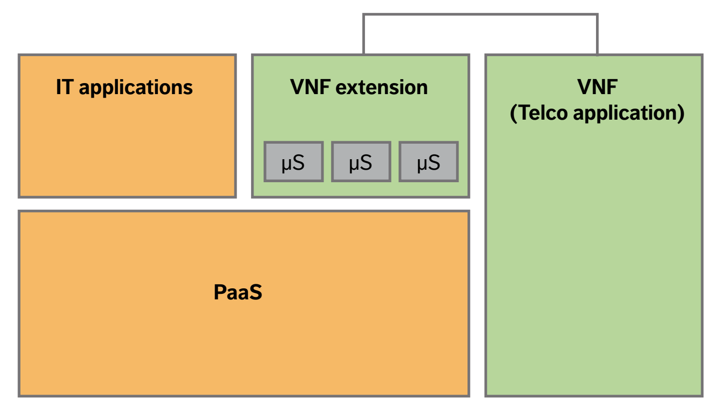

Figure 3
Extension of telco applications in PaaS

adapters, and translators can be applied in this process [3]. Such new satellite microservices need to be managed through the application to avoid fragmentation – which would incur additional costs.

The proposed approach brings all the benefits of PaaS and microservices without the transformation cost. It also has the advantage of gradually allowing teams to experience new technologies and new ways of working. In case of non-adoption owing to a performance degradation or a mismatch with the way of working in the organization, reverting to the original development method would have little impact. Going to market with such an approach would also create the opportunity for early feedback from the customer.

The next steps

Telco applications deployed in PaaS – in particular those specifically designed for PaaS environments – can benefit from shorter development, deployment, and testing cycles than the traditional software stack. Telco PaaS is characterized by short time to market and rapid innovation, enabling developers to focus on the business logic of their applications and building lightweight software modules that maximize reuse of platform features and available services.

Application scalability and the introduction of new technologies are both facilitated by the independent life cycles of application components and the services they use. PaaS increases the portability of the applications to several IaaS solutions, and thus helps reduce the number of cloud execution platforms that need to be supported.

Improving PaaS solutions to support the additional requirements of telco applications can bring benefits to users outside the telco domain, making PaaS more suitable for other domains that have stringent requirements in terms of, say, latency and security – such as in the banking or medical sectors.

However, as outlined, a number of gaps and challenges need to be overcome before the combination of PaaS and microservices can reveal its full potential. Issues like the maturity of existing PaaS platforms, the possibility to perform troubleshooting in a highly distributed system, testing that involves a large number of independent services, the ability to predict latency (or maintain it within set limits), management support, and closing the gaps relating to telco-grade applications are just some of the challenges that need to be overcome.

Designing a microservice-based application for PaaS requires applications to follow a number of recommended design patterns to achieve a robust, functional application. PaaS can be leveraged without a full redesign and rewrite of the current telco applications, extending an existing application by deploying new features as extensions in PaaS, while the rest of the application remains outside.

The potential benefits of a mini- or microservice-based architecture deployed in a PaaS environment are significant. As such, Ericsson will follow the steps along the path to telco-grade PaaS. ✪

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4G/5G RAN architecture

HOW A SPLIT CAN MAKE THE DIFFERENCE

As 5G evolves, many innovative services like extreme mobile broadband and long-range massive MTC will come into play. In line with the evolution of 4G and the introduction of 5G, RAN architecture is undergoing a transformation: to increase deployment flexibility and network dynamicity, enabling networks to meet increasing performance requirements, while at the same time keeping a lid on the total cost of ownership. The proposed future-proof software-configurable split architecture will be able to support new services, deployed on general-purpose and specialized hardware, with functions ideally placed to maximize scalability, spectrum, and energy efficiency, all while supporting the concept of network slicing.

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THE REQUIREMENTS created by extreme MBB, the IoT, and massive MTC call for an alternative to today's deployment architectures. The changes to architecture include the capability to place selected functions closer to the network edge, for example, and the ability to increase RAN resilience. Cost is naturally a factor, as spectrum availability and site infrastructure continue to dominate operator expenditure for wide-area systems. The evolution of RAN architecture therefore needs to include measures for enhanced spectrum efficiency that are harmonized with other improvements in the areas of hardware performance and energy efficiency.

In light of these cost and performance requirements, a number of capabilities are shaping the evolution path of RAN architecture:

Seamless Radio Resource Management

The best combination of any radio beam within reach of a user should be used for connectivity across all access network technologies, antenna points, and sites. This capability will be achieved by applying carrier aggregation, dual connectivity, CoMP, and a number of MIMO and beamforming schemes.

Functional split

Some 5G requirements – such as ultra-low latency and ultra-high throughput – require highly flexible RAN architecture and topology. This will be enabled by splitting RAN functions, including the separation of the user plane (UP) and the control plane (CP) in higher layers.

Dynamic and software-defined RAN

The capability to configure, scale, and reconfigure logical nodes through software commands enables the RAN to dynamically adjust to changing traffic conditions, hardware faults, as well as new service requirements. This capability will be achieved by separating out logical nodes suitable for virtualization (on a GPP) and designing functions that require specialized hardware to be dynamically (re)configurable on an SPP.

> “THE CAPABILITY TO CONFIGURE, SCALE, AND RECONFIGURE LOGICAL NODES THROUGH SOFTWARE COMMANDS ENABLES THE RAN TO DYNAMICALLY ADJUST TO CHANGING TRAFFIC”

Deployment flexibility

Deployment flexibility enables an operator to deploy and configure the RAN with maximum spectrum efficiency and service performance regardless of the site topology, transport network characteristics, and spectrum scenario.

This is achieved through a correct split of the RAN architecture into logical nodes, combined with the future-proof freedom to deploy each node type in the sites that are most appropriate given the physical topology and service requirements.

The process to reach the target architecture with the correct split involves a number of steps:

- determining the logical functions that comprise 5G RAN on a level below 3GPP
- identifying the latency-critical functions that need to be placed within a few TTIs to antenna elements
- identifying which functions have more relaxed latency requirements

Terms and abbreviations

BPF – baseband processing function | **CO** – central office | **CoMP** – coordinated multipoint | **CP** – control plane | **CPRI** – Common Public Radio Interface | **C-RAN** – cloud RAN | **DL** – downlink | **EPC** – Evolved Packet Core | **E2E** – end-to-end | **GPP** – general purpose processor | **HARQ** – hybrid automatic repeat request | **IoT** – Internet of Things | **MAC** – media access controller | **MBB** – mobile broadband | **MIMO** – multiple-input, multiple-output | **MTC** – machine-type communications | **NFV** – Network Functions Virtualization | **NR** – next generation RAT | **OSS** – operations support systems | **PDCP** – packet data convergence protocol | **PDU** – protocol data unit | **PGW** – packet data network gateway | **PHY** – physical interface transceiver | **PPF** – packet processing function | **RAT** – radio-access technology | **RCF** – radio control function | **RDC** – regional data center | **RF** – radio function | **RLC** – Radio Link Control | **RRM** – Radio Resource Management | **SON** – self-organizing networks | **SPP** – special purpose processor | **S-RRM** – server RRM | **TTI** – time transmit interval | **UE** – user equipment | **UL** – uplink | **UP** – user plane | **U-RRM** – user RRM | **VNF** – Virtualized Network Function

» determining where to place anchor points for soft combining, carrier aggregation, and dual connectivity – among the user-plane functions
» identifying which nodes can be implemented as VNFs

This process is illustrated in *Figures 1*, *2*, and *3*. *Figure 1* shows the 4G/5G logical architecture at a level below 3GPP; *Figure 2* shows today's 4G split into an RU and a DU; and *Figure 3* shows the target split architecture. Throughout the process, and as a result of the functional decomposition, new inter-node interfaces emerge, whose characteristics need to be taken into consideration to ensure that the underlying transport network can support the various deployment scenarios.

The logical 4G/5G RAN architecture

The external interfaces of the RAN domain (except to the OSS) are standardized under 3GPP, as is the functional behavior of the RAN domain as a whole. Below the high-level specification, 3GPP leaves room for innovation to enhance the network with RAN-internal value-add features – a flexibility that has over a number of years resulted in continuous improvement in many areas, including spectrum efficiency (in the form of scheduling algorithms, power control algorithms, and various RRM features), energy efficiency, and enhancements to service characteristics such as lower latencies. To determine the optimal architectural split, however, the RAN architecture needs to be examined with a finer level of granularity than that offered by 3GPP.

RAN anchor points

Figure 1 illustrates the logical RAN architecture, which for the purposes of simplification shows the UL and DL instances of each function combined, and the solid lines indicate user plane functions. In the downlink, PDUs enter the RAN domain over the S1-U interface (right) and are delivered to devices (on the left) over the radio interface. The multipath-handling function is the anchor point for

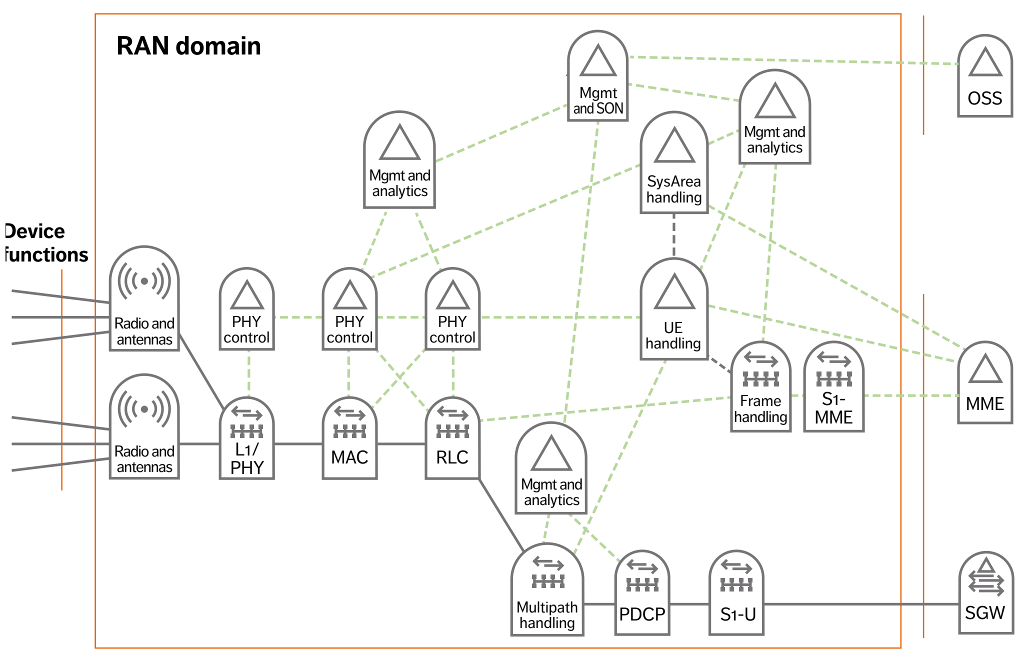


Figure 1
The logical 4G/5G RAN architecture one level below 3GPP

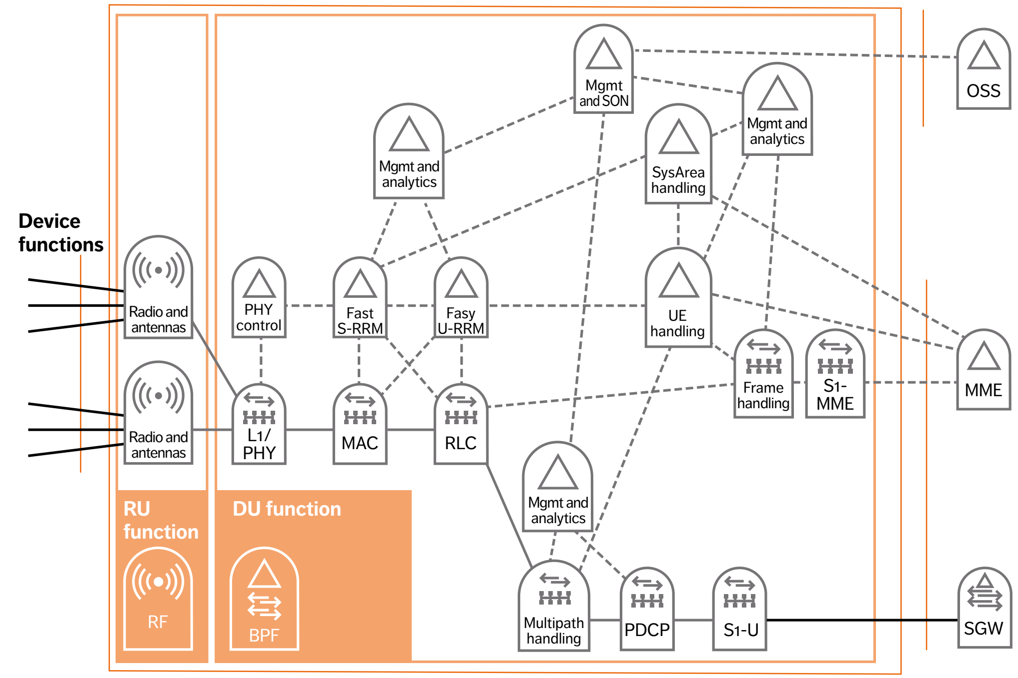


Figure 2
Logical RAN architecture – present split into an RU function and a DU function

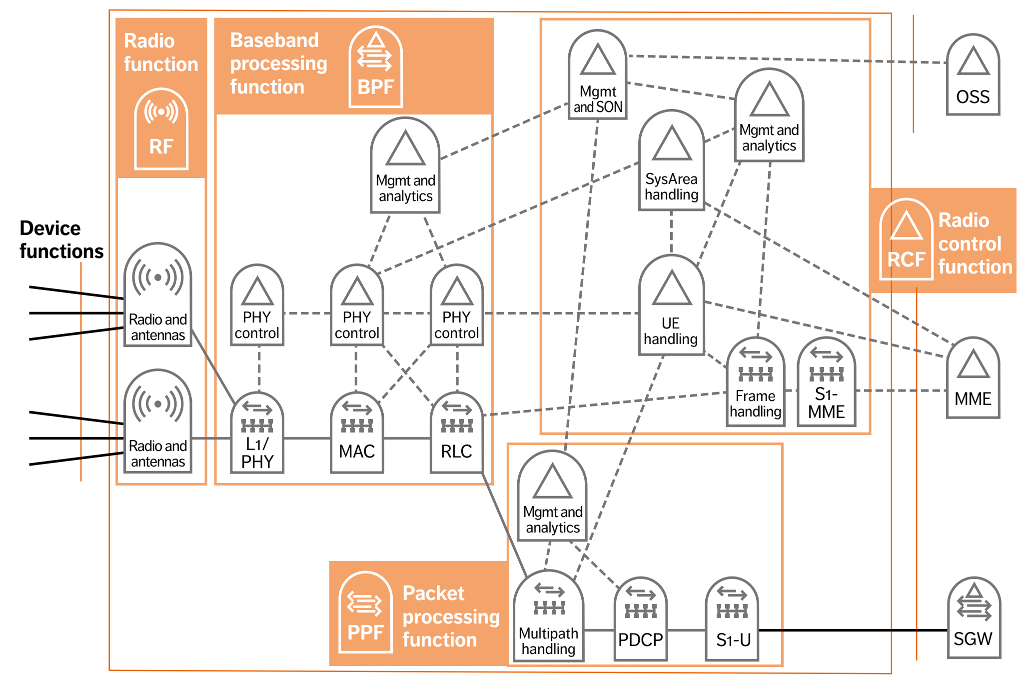


Figure 3
Logical RAN architecture – target split into the functions RF, BPF, PPF, and RCF

VIRTUALIZE, SPLIT, AND REORGANIZE ✱

dual connectivity, which schedules individual PDCP PDUs in the same user-data stream to different RLC instances, possibly on different RATs. In this way, a single UE can simultaneously receive and send data over different radio channels – for example, one NR and one LTE channel – that are connected to different sites. The MAC function is the anchor point for carrier aggregation, which schedules MAC PDUs to each user over a multitude of 4G or 5G carriers. The MAC function handles COMP and multi-beam transmissions. In the uplink, the L1/PHY function performs soft combining, the MAC function aggregates UL data in carrier aggregation, and multipath handling aggregates data received from dual connectivity UL data streams.

The HARQ loop

3GPP specifies retransmission periods and response times at the radio level between the UE and the network as the HARQ loop, which includes: the air-interface transmission time, completion times for RF-L1/PHY-MAC functions, and RF-L1/PHY-MAC functions in the UE.

The loop results in a standardized round-trip latency budget of 3ms for LTE, and down to 200µs for NR. The RAN functions participating in the loop work synchronously with the air-interface TTI, while the PDCP and multipath handling function feed and receive packets traveling to and from the RLC layer asynchronously – which has implications for the split architecture.

Control plane functions

Runtime control functions can generally be divided into three categories, depending on whether they: act on a per-user basis (U-RRM), control spectrum on a system level (S-RRM), or manage infrastructure and other common resources.

The U-RRM functions include measurement reporting, selection of modulation and coding schemes, per-UE bearer handling, and handover execution. The functions in fast U-RRM act within the HARQ loop and serve the scheduler with processed real-time per-UE information. In contrast, the U-RRM function (UE-handling) works on a time scale of 10ms and above, including bearer handling, per-UE policy handling, handoff control, and more.

Functions that control spectrum on the system level include radio scheduling, distribution of the power budget across active UEs, and system-initiated load-sharing handovers. The fast S-RRM operating within the HARQ loop is responsible for radio scheduling and functions in tight coordination with the MAC and RLC. System-area handlers – such as load sharing, system information control, and dual-connectivity control – control spectrum on a 10ms time scale, or slower.

Functions that control infrastructure and common resources – other than spectrum – include handling of transport, connectivity, hardware, and energy. By allowing the control functions for spectrum, transport, infrastructure, and connectivity to interact, a holistic control system for RAN resources can be built.

Interface characteristics

The RAN HARQ loop time budget of three TTIs is divided into time for processing, and time for signals and data to traverse the various inter-function interfaces. The less time spent on interface signaling, the more time is available for processing, which translates into lower cost for hardware and for energy consumption. To minimize signaling latency, and thus maximize hardware efficiency, the MAC, RLC, and fast-U/S-RRM functions should run on the same hardware instance. As traffic moves to the right in *Figure 4*, the requirement on interface latency gradually relaxes.

The CPRI scales with effective carrier bandwidth and the number of antenna elements. An inter-site CPRI interface with joint combining at the central office (CO) can, in many deployments, result in a gain in uplink spectrum. For LTE, the CPRI can be inter-site (a few Gbps), but in NR – with wider carriers and more antenna elements – an inter-site CPRI would be challenging from both a latency and bandwidth perspective.

The interface between the RLC and the multipath handler scales with user data and has a latency tolerance in the order of several milliseconds. This interface is limited by the performance of the dual connectivity feature, which degrades gracefully as the interface latency increases. This interface can, therefore, either be node internal or a networked interface between nodes and even between sites that are not more than 3-5ms apart.

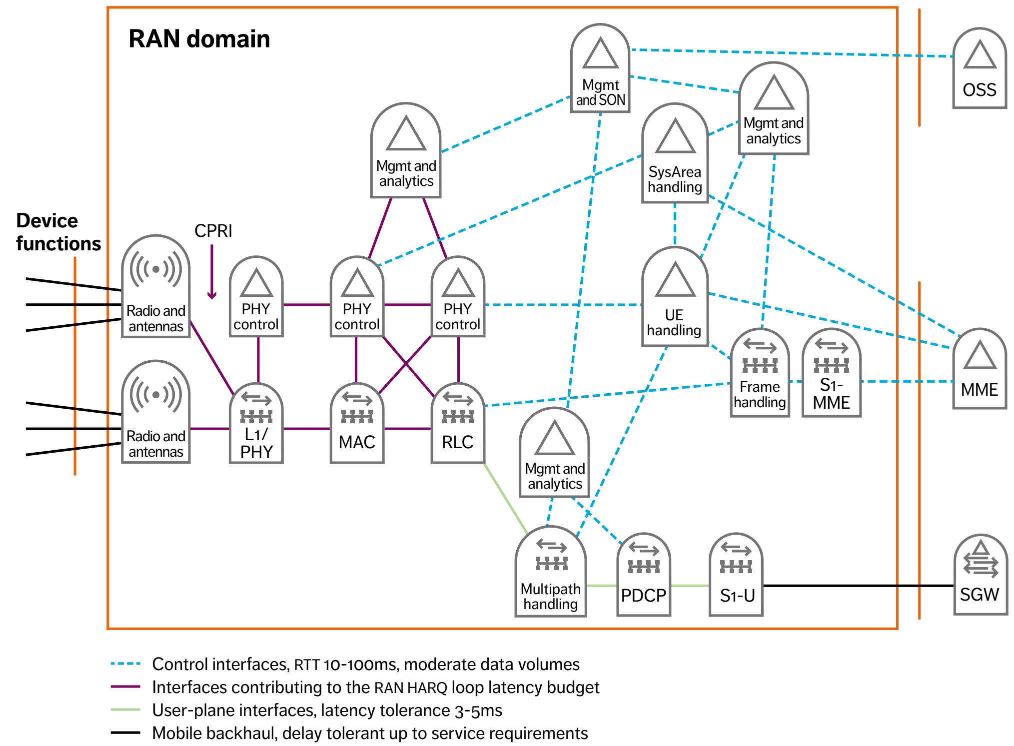


Figure 4
The logical interfaces in the RAN architecture and their characteristics requirements

The remaining interfaces in Figure 4 carry either slow control data (indicated by the blue lines) or user data between the RAN and the EPC (S1-U interface). At over 10ms, the latency requirements on these interfaces are quite relaxed.

Hardware requirements

Control functions that are asynchronous to the radio interface tend to be suitable for virtualization and VNF deployments, as they are transaction based and do not involve heavy packet processing.

On the other hand, functions like multipath handling, PDCP, and S1-U termination involve packet processing (encapsulation, header reading/creation, encryption/decryption, and routing) and can be challenging to virtualize. However, if the underlying hardware contains ciphering offload and packet-processing accelerators, virtualization is possible without performance degradation, and so these functions can be virtualized and deployed in an NFV environment.

Most RAN processing cycles occur in the HARQ synchronous functions. Tasks like uplink radio decoding and scheduling are, for example, highly processing intense. And so, the more processing that can be carried out in the uplink decoding, the better the uplink sensitivity, and the more processing that can be allocated to the scheduler, the better the use of spectrum in both the uplink and the downlink.

Reducing processing in the HARQ synchronous

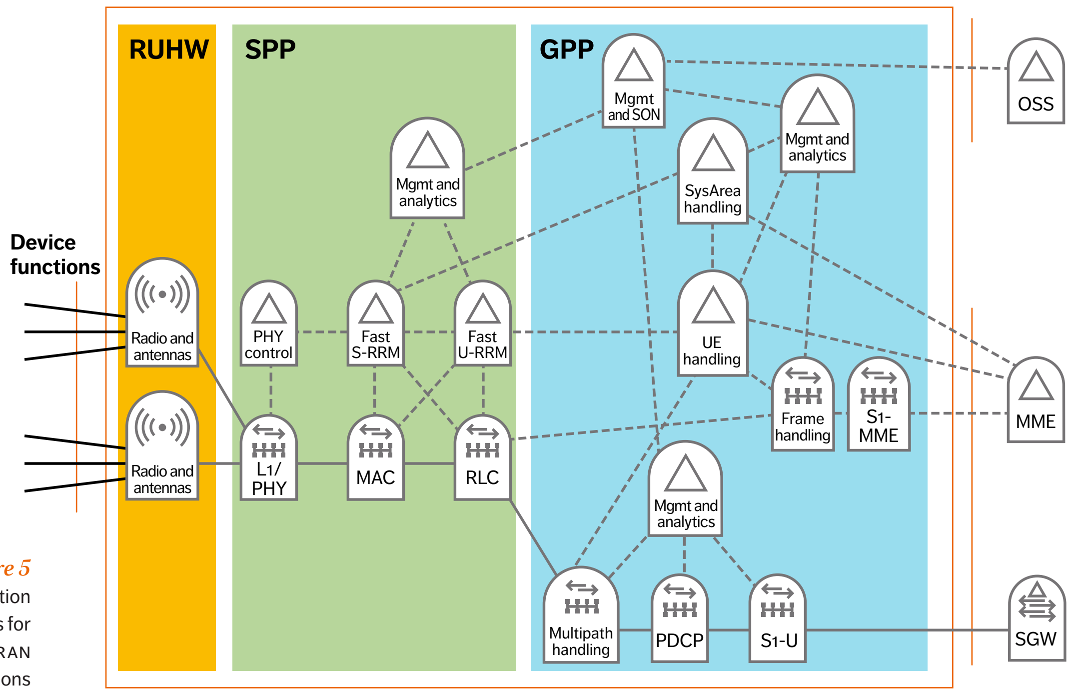


Figure 5
Preferred execution environments for the various RAN functions

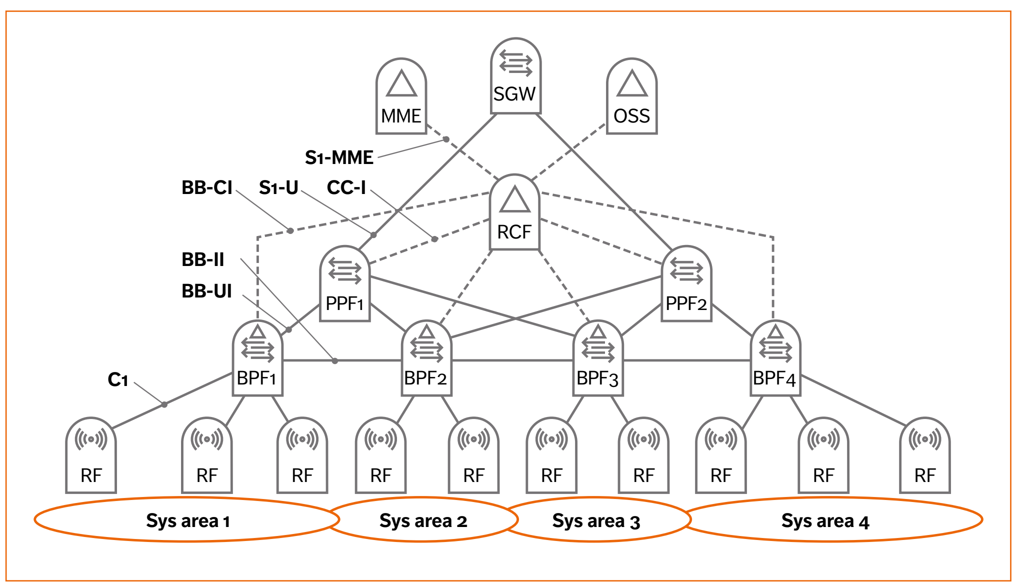


Figure 6
Hierarchy, instantiation, and inter-node interfaces in the RAN split architecture

functions is not a good idea if spectrum efficiency is to be maintained or improved. Special-purpose multi-core hardware is best suited to this type of processing, as its price-performance ratio is presently five times that of single-core hardware. And so, HARQ synchronous functions are likely to continue to run on special purpose processor (SPP) hardware for at least one or two more generations of RAN hardware.

To avoid flow control issues between the MAC and the RLC – and given the level of interaction between the scheduler and RLC – the MAC, RLC, and the fast RRM should run on the same hardware instance. The resulting hardware environment is shown in *Figure 5*.

The functions on the right in the illustration (blue area) run on general purpose processors (GPPs) that include hardware accelerators and ciphering offload for multipath handling and PDCP. The functions in the middle (green area) run on SPPs with multi-core hardware suitable for supporting functions in the HARQ loop, and the radio hardware is on the left (yellow area).

Resulting split architecture

Based on function and interface characteristics, preferred execution environment, and spectrum efficiency, the target functional composition, which is shown in *Figure 6*, includes the following logical RAN nodes:

Packet processing function – PPF

The PPF, which is suitable for virtualization, contains user-plane functions that are asynchronous to the HARQ loop, and includes the PDCP layer – such as encryption – and the multipath handling function for the dual connectivity anchor point and data scheduling.

Baseband processing function – BPF

Given the stringent requirements for spectrum efficiency, the BPF benefits from being placed on an SPP. The BPF includes user-plane functions that are synchronous to the HARQ loop, including the RLC, MAC, and L1/PHY, and it is also the anchor point for carrier aggregation (MAC) and soft combining (L1/PHY). The BPF contains the per-TTI RRM (fast radio scheduler), and is also responsible for the COMP, for the selection of the MIMO scheme, and for beam and antenna elements.

Radio function – RF

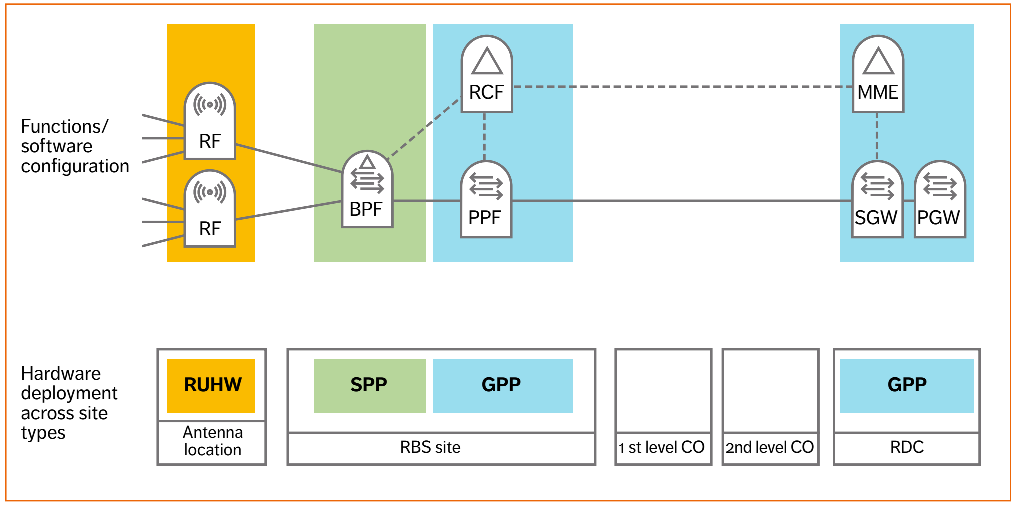
The RF requires special radio hardware and includes functions such as modulation, D/A conversion, filtering, and signal amplification.

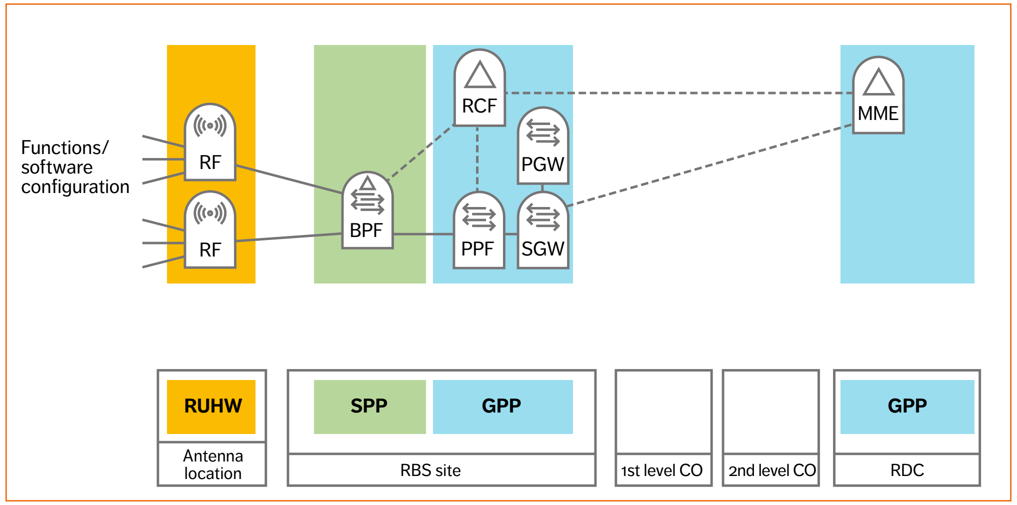
Radio control function – RCF

The RCF handles load sharing among system areas and different radio technologies, as well as the use of policies to control the schedulers in the BPFs and PPFs. At the user and bearer level, the RCF negotiates QoS and other policies with other domains, and is responsible for the associated SLA enforcement in the RAN. The RCF controls the overall RAN performance relative to the service requirement, creates and manages analytics data, and is responsible for the RAN SON functions. Like the PPF, the RCF is suitable for virtualization.

The logical interfaces are:

- C1 – an evolution of the CPRI interface, this interface scales with effective carrier bandwidth (carrier bandwidth × number of antenna streams) with a latency requirement of around one TTI (1ms for LTE and down to 67µs for NR).
- BB-UI – the user-plane interface between the PPF and the BPF, it carries PDCP PDUs, which scale according to the amount of user data sent by the BPF instance to the system area.
- BB-II – the interface between two BPF instances, it carries user data for scenarios that use inter-BPF carrier aggregation (carrier aggregation over two carriers controlled by two different BPF instances). This interface also carries control data for the inter-BPF COMP, scales in line with the amount of user data transmitted, and the latency requirement is the same as for C1.
- BB-CI – the control-plane interface for the BPFs, which carries control and analytics data from the BPF to the RCF. This interface primarily scales with the volume of analytics data, and at 10ms or more, the latency requirements are quite relaxed.
- CC-I – the control-plane interface for the PPFs, which





Figures 7a and 7b
Deployment of hardware and nodes across site types in a classic main-remote deployment valid for both LTE and NR

carries control and analytics data from the PPF to the RCF. Like BB-C1, this interface scales primarily according to the volume of analytics data transmitted, and has relaxed latency requirements of 10ms or more.

In a deployment, each function (RF, BPF, PPF, and RFC) is instantiated. An instance of the radio functions will be associated with a number of antenna elements at an antenna site, and a set of *n* RF instances are connected to one instance of the BPF.

Each antenna element (RF) is associated with one BPF. Hence, a BPF instance handles the cells corresponding to the RF antenna elements it is associated with. The set of cells under the control of a BPF instance is referred to as a system area. By definition, one BPF instance handles the radio transmission and reception of traffic in its system area. Within its system area, a BPF instance also controls beamforming, power, spectrum, scheduling, load sharing, and fast U/S-RRM.

Mobility within a system area is hidden under the BPF and not visible to the PPF or the EPC. A multitude of BPF instances are connected to one instance of the PPF. One PPF instance is therefore associated with a large number of BPFs, as well as the traffic sent to and by users in a large set of system areas. As the S1-U terminates in the PPF, a user can move around within this set of system areas or cells without causing an S1 or X2 handover. Each instance of the RCF can handle a small or a large set of PPFs – and all the associated BPFs. In this way, the RCF can keep a holistic view of an area that is just a single cell, up to an area consisting of thousands of cells. With this architecture, RRM coordination and spectrum efficiency within a system area can be maximized, using the full suite of RRM features available, which lays the foundation for the application of future innovations and RRM technologies.

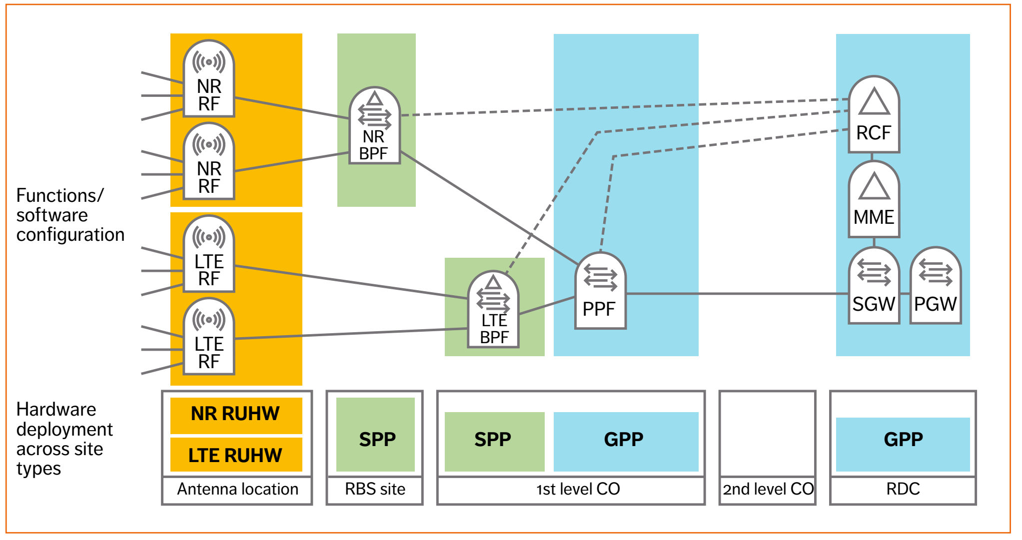
Deployment alternatives and examples

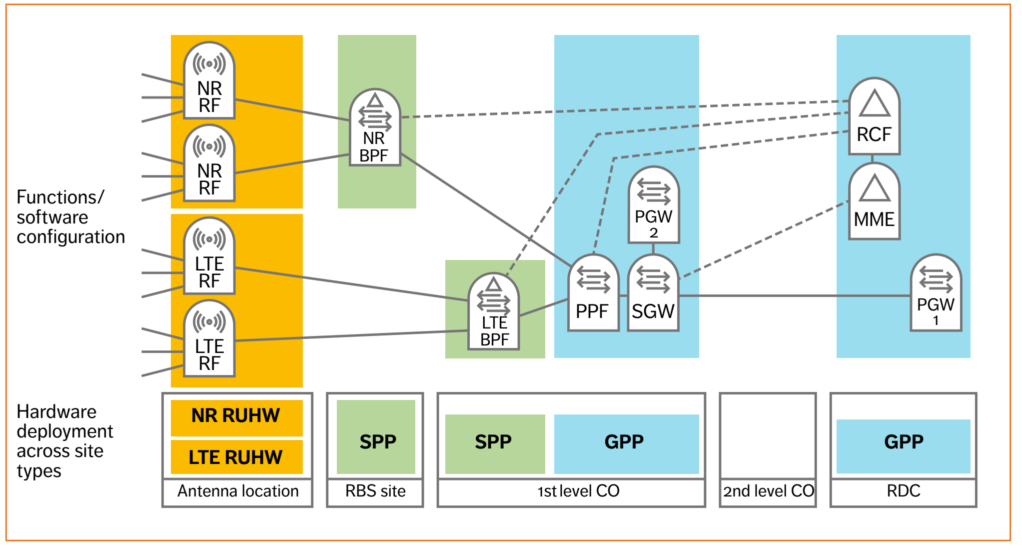
Having defined the split RAN architecture, it is possible to describe some of the deployment alternatives it supports and the associated dynamics of the software-defined radio network. *Figures 7a* and *7b* show a distributed main-remote macro site. The hardware deployed at each site is shown in the bottom half of each figure, while the top part shows the configuration of network functions across that hardware. The hardware deployment is semi-static, but the network functions are (re)configured using commands or machine instructions, and are therefore software defined.

The antenna location contains radio unit hardware that hosts the RFs. In the distributed main-remote deployment, a fiber link connects the antenna location to the RBS main site over a CPRI (C1 interface). The RBS site is configured with an SPP for the BPF, and a GPP for the PPF and RCF, and is connected over S1 mobile backhaul transport to EPC gateways residing in a more centrally located regional data center (RDC). In addition to providing an execution environment for the PPF and RCF, the GPP hardware offers a virtualization environment for VNFs and applications.

This deployment enables computing at the mobile edge for selected applications, users, or data streams to be moved to the RBS site by means of a system reconfiguration – or even automatically based on policies and traffic triggers. In Figure 7b, a virtual packet gateway – deployed as a VNF – is located at the RBS site running in the virtualization environment of the GPP. While Figures 7a and 7b describe the same hardware installations, the configuration of 7b supports computing at the mobile edge in the RBS site – due to the ability to break out traffic in the local packet gateway function. In 7a, all the traffic is backhauled to the gateway in the RDC. As the hardware deployments in 7a and 7b are the same, the radio-network architecture difference is created through dynamic software commands. In this way, the network architecture can adapt according to policies and varying traffic conditions, either through software configuration or automatically (automation and SON) – which is the basis for software-defined RAN.

A second example, illustrated in *Figure 8*, shows a typical LTE C-RAN deployment extended with main-remote NR. As *Figure 8a* illustrates, the radio unit hardware is located at the antenna location, while the SPP and GPP are deployed in the central office sites. The SPP hardware runs one or several instances of the BPF, and the GPP provides the NFV





Figures 8a and 8b
C-RAN deployment with (instance *a*) a centralized packet gateway and (instance *b*) a second instance of the PGW distributed to the CO site for local breakout of selected traffic

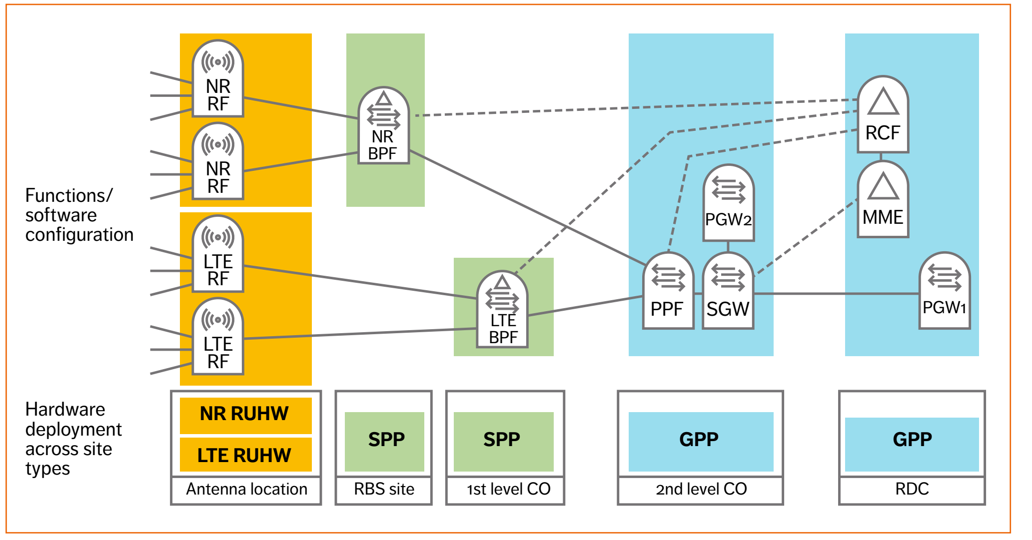


Figure 9
Flexible C-RAN architecture second-level CO for better dual connectivity performance

environment for the PPF and the RCF. Compared with the distributed macro deployment shown in Figure 7a, the LTE BPF in the C-RAN deployment holds a more centralized position. Each instance of the BPF covers more antenna locations, which results in high spectrum efficiency, as the BPF can use the complete set of fast RRM features — including joint combining, multipoint transmission, and coordinated scheduling — across all the antenna points in the system area covered by the first-level central office site.

The C-RAN deployment requires high-bandwidth fiber for the C1 interface between the BPF and the RF (CPRI fronthaul). Like the distributed macro architecture of 7b, the C-RAN deployment offers a GPP environment for VNFs and applications in the central office. The GPP may be part of a distributed data center, in which case, the PPF and RCF are deployed as VNFs, while the SPP is deployed as standalone hardware. Alternatively, the SPP can be part of the distributed data center, providing specialized hardware for the BPF — similar to the way other types of data-center specialized hardware can be used by applications with special needs, such as packet processing or firewalling. In this way, next generation central offices can be turned into a combination of mobile-edge sites and baseband hotels.

The flexible C-RAN architecture illustrated in *Figure 9* supports more centralized deployments compared with the C-RAN configuration shown in Figure 8. The GPP in the flexible C-RAN architecture is provisioned at the next level in the hierarchy, moving the PPF and hence the dual-connectivity anchor point to a more central position, which enables smooth dual connectivity mobility. Spectrum efficiency is the same or slightly improved, the number of distributed data centers drops, and the mobile edge is slightly more centralized. The transport requirements of both C-RAN configurations are similar.

Figure 10 shows an on-premises architecture that is fully self-contained using pico base stations and an on-premises data center hub that stores content and carries out processing locally. In this case, all four RAN nodes (RF, BPF, PPF, and RCF) are integrated onto the same chip.

The ideal split architecture contains pools of hardware (an SPP and GPP) strategically deployed in selected RBS and CO sites. Instances of the three function types BPF, PPF, and RCF – and any VNFS in the mobile network – are dynamically created, modified, scaled, and terminated based on need and operator policies. Building mobile networks in this software-defined way results in an architecture that:

- is reliable and resilient – as it enables functions to relocate following a hardware, site, or link failure
- is flexible and energy efficient – as it automatically adapts to peaks and troughs in traffic patterns and service usage
- reduces time to market for new services and network features
- can maximize spectrum efficiency (minimize cost for spectrum and sites) given the geography and site topology of the network, through its flexibility to dynamically distribute the hub points for joint combining, multipoint transmission, COMP, carrier aggregation, and dual connectivity

Conclusions

The design of the 4G/5G split RAN architecture focuses on increased spectrum efficiency, full deployment flexibility, and elasticity; processing is carried out where resources are available and needed. The split RAN architecture consists of the two user-plane network functions: a packet processing function (PPF) and a baseband processing function (BPF), together with the antenna-near radio function (RF), and the control-plane radio control function (RCF).

The PPFS and RCFS can each be deployed either in classical pre-integrated nodes or in fully virtualized environments as VNFS or any combination thereof. Both functions are suitable for virtualization with existing technology, with benefits to the PPF brought by packet accelerators and ciphering support in the underlying hardware.

In principle, the BPF can also be virtualized. However, special-purpose hardware is still about five to seven times more efficient than general-purpose hardware for the type of processing the BPF performs, and so it is expected that the BPF will be deployed on an SPP for one or two more generations of hardware.

The RCF takes holistic responsibility for Radio Resource Management, RAN analytics and SON, it maintains policy and bearer information, and interworks with non-RAN domains such as the EPC and resource orchestration layers. The RCF can be centralized or distributed in a closed on-premises factory network, for example. Deploying the user-plane BPF (processing synchronous to the TTI) and PPF (asynchronous packet processing) can be achieved in a variety of ways, as long as the BPF is within one to two TTIS from the antenna points. The PPF, on the other hand, can be more centralized, with a distance of up to 5-7ms from the radio functions. And so user-plane functions can be deployed to match service requirements, and maximize spectrum efficiency according to the spectrum, transport, and site availability, as well as the particular local geography.

The split architecture results in the necessary scaling dimensions to support 5G use cases and traffic structures in a cost-efficient way. Its flexibility and decoupling of hardware from software enables a software-defined elastic resilient RAN. It also guarantees that RAN architecture is future-proof. As an evolution of 4G RAN, the split can be gradually introduced in line with business needs. ✪

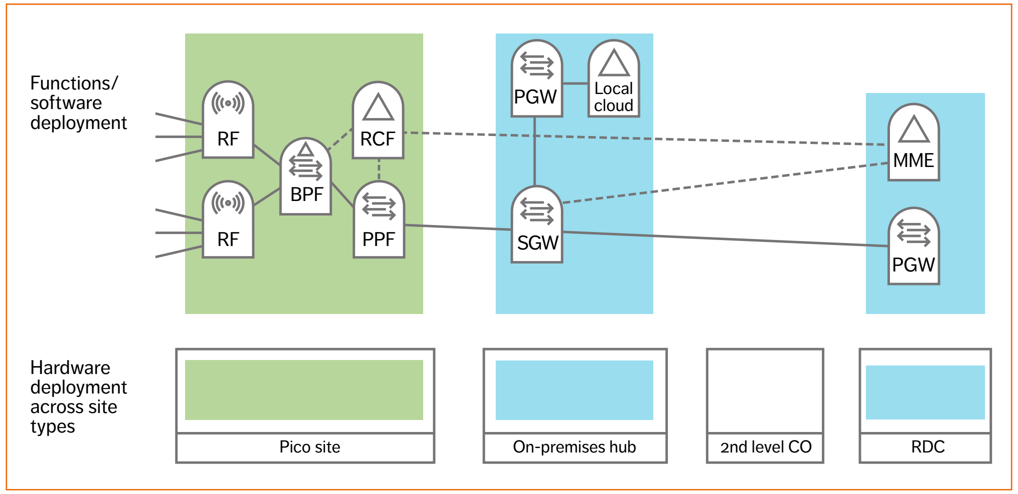


Figure 10
Factory deployment examples using pico 4G/5G base stations (green) and with on-premises breakout possibility

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AUTOMATION EVERYWHERE ✱

Cloud robotics

5G PAVES THE WAY FOR MASS-MARKET AUTOMATION

Robots, robotics, and system automation have shifted from the floor of the research lab to becoming a crucial cost, time, and energy-saving element of modern industry. Smart robots and their control systems have enabled entire processes, like vehicle assembly, to be carried out automatically. By adding mobility to the mix, the possibilities to include system automation in almost any process in almost any industry increase dramatically. But there is a challenge. How do you build smart robotic systems that are affordable? The answer: cloud robotics enabled by 5G.

MARZIO PULERI
ROBERTO SABELLA
AFIF OSSEIRAN

THE IMPACT of robots and robotics has been greatest in the manufacturing industry, particularly on assembly lines and in hazardous environments. Traditionally, robots have been designed to perform repetitive pick-and-place tasks, to carry out operations that require a high degree of precision, as well as doing hazardous jobs like welding and cutting. But things are changing... Robots are beginning to appear in all industries, and are being used to carry out a wide variety of tasks.

■ Smarter than ever, modern robots are capable of adapting to changing conditions. The current drawback of the smart robot, however, is the massive amount of intelligence it needs to function correctly – resulting in complex machines and control systems. Coupled with the fact that smart robots tend to be built with all the hardware and software they need, these automatons tend to be costly – a factor that is slowing down development cycles and hampering uptake in new sectors.

Cloud robotics aims to change this by putting systems intelligence in the cloud and simplified robotics on the ground.

Within this model, mobile technology plays the key enabler role – connecting the cloud-based system to the robots and controllers in a system. And, it is high-performance mobility that will provide the latency and bandwidth needed to support system stability and information exchange, which in turn facilitates the building of sophisticated, yet affordable, robotic systems.

Within mobile, radio technologies will provide the wanted level of performance, and so it is the capabilities of 4G and 5G radio systems that will enable 5G cloud robotics and facilitate the uptake of robotics in new applications.

Leading industries

Naturally, manufacturing is one of the industries taking the lead when it comes to cloud robotics, but others sectors like health care, transportation, and consumer services are among the forerunners contributing to the evolution of robotics.

In manufacturing, cloud robotics will improve the performance of a production plant, for example, through preventive maintenance, and support advanced lean manufacturing. By storing intelligence in the cloud, it is possible to increase the level of automation of a system, regulate on-the-fly processes, and prevent malfunction and faults.

In the health care sector emerging applications include robot-assisted remote patient care, automation and optimization of hospital logistics, and cloud-based medical service robots. As is the case in most industries, initial solutions will take care of simple tasks, with development leading to more complex and demanding applications, such as remote surgery, further down the line.

Driver-assisted, autonomous, and semi-autonomous vehicles, automatic transportation for the disabled, and unmanned delivery services are just some of the emerging applications in transportation. And in consumer services, domestic robots, automated wheelchairs, personal mobility assistants, and leisure robots exemplify the types of solutions on the design table.

> “4G AND 5G RADIO SYSTEMS WILL ENABLE CLOUD ROBOTICS AND FACILITATE THE UPTAKE OF ROBOTICS IN NEW APPLICATIONS”

The IoT and the fourth industrial revolution

While some industries might be leading development, all sectors have a part to play in the evolution of cloud robotics, as they deploy robots to carry out a wide variety of tasks [1]. Here are just a few examples:

- farming: spraying, fertilizing, harvesting, and milking
- construction: assembly, dispensing, laying, and welding
- utilities: remote control, inspection, and repair

As a key element of the IoT and an enabler of the fourth industrial revolution, robotics and robots will vary greatly in terms of application, agility, complexity, cost, and technical capability. But most – if not all – robots will require wireless connectivity to the cloud and wider analytics ecosystem [2].

Of all the industries that will shape robotics, industrial IoT is expected to have the greatest impact in the short term. And so, to best exemplify the business model for cloud-connected robots, this article focuses on industrial IoT.

Terms and abbreviations
AMCL – Adaptive Monte Carlo Localization | **CAN** – controller area network | **COTS** – commercial off-the-shelf | **DWA** – dynamic window approach | **NAT** – network address translation | **ROS** – robot operating system | **UGV** – unmanned ground vehicle | **VM** – virtual machine

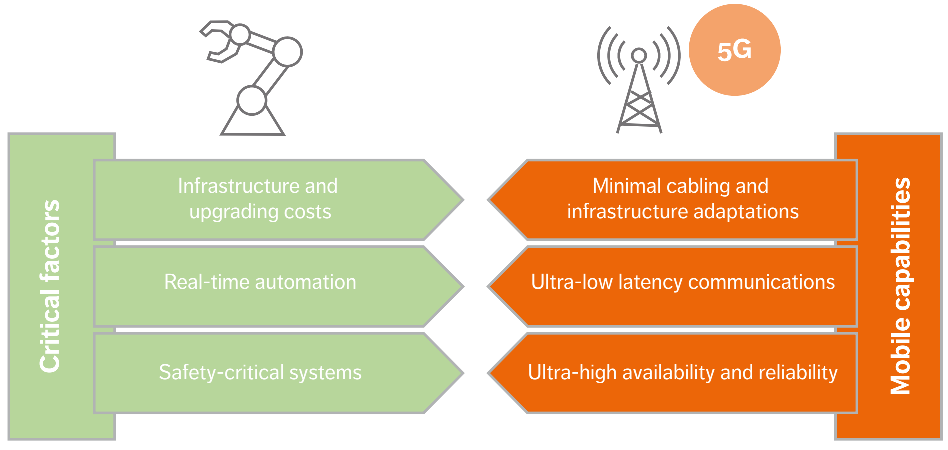

Figure 1
Mobile connectivity offers flexibility

Stakeholder roles

Within the cloud-robotics domain there are a number of different roles or players:

- the raw robot provider who builds COTS robots and customized machines based on requirements set by a manufacturer
- the local access provider or ICT player who connects the on-site robots (in a factory, on a construction site, or on a farm, for example) to a local gateway or hub
- the local systems integrator or ICT player who integrates (wired and wireless) access, data processing, cloud systems, security procedures, and manufacturing execution systems (MESs) infrastructure
- the global network access and cloud provider or ICT player that extends connectivity to a global macro network outside the site
- the industrial IoT ICT system provider or ICT player that automates, digitalizes, and connects the full industrial cycle, from raw components to product servicing, and recycling

Ecosystem

Given the potentially massive range of use cases, the business prospects for cloud-based robotics look promising. However, as is often the case with opportunity, cloud robotics presents certain challenges. Since the one-size-fits-all business model doesn't work, a shared platform for common functions is needed; which can then be tailored to particular applications by building use-case-specific modules on top.

Spectrum

The business models that apply to a given use case will ultimately depend on the spectrum sharing regime — licensed or unlicensed — which will in turn set the boundaries for the performance of wireless cloud-robotic technology solutions.

Security

Information security needs differ from one industry to the next. For example, the demands posed by industrial IoT applications are more stringent than they are for consumer products. For industrial IoT, the multiple actors in the supply chain need to be able to share data in a controlled and secure environment. To be able to address the security requirements for most industrial IoT use cases, information sharing needs to ensure:

> “THE FLEXIBILITY OFFERED BY WIRELESS CONNECTIVITY ENSURES THAT MODIFICATION COSTS CAN BE KEPT TO A MINIMUM”

- the ability to control and limit data access
- data is available to all parties in the value chain at all times
- that all parties can trust the data being shared

The benefits of wireless connectivity

When deploying a cloud-based robotic system, constant and reliable connectivity for each robot is essential, irrespective of the type of robot, the tasks it carries out, or the environment in which it operates. Since with wireless, there is no need for complex and inflexible wireline connections, it is the obvious solution to provide communication to systems that rely on mobile robots moving along set paths or following on-the-fly directions. But even for systems like assembly lines where fixed robots are the primary automatons, shifting to wireless connectivity significantly simplifies the overall solution — limiting cabling needs to just power supply. As *Figure 1* shows, what wireless connectivity achieves is flexibility. And so, when changes need to be made, for example, to the layout of a manufacturing plant to adapt to a new production process, or to add new robots, the flexibility offered by wireless connectivity ensures that modification costs can be kept to a minimum, which in turns offers greater flexibility in decision-making at the business level.

Selecting the best wireless connectivity technology naturally depends on the operating environment of the given scenario. Without diving into specific use cases, NB-IoT, Wi-Fi, LTE, and soon 5G technologies loosely apply to different categories of applications, as follows:

NB-IoT technology can be applied to industrial applications like process monitoring in chemical plants. This technology is attractive in terms of battery life, cost, and coverage, but it is not suitable for applications like remote control of machinery that demand high bitrate and low latency.

Wi-Fi is a good option for scenarios where robots are mostly fixed and placed in a well-defined context. Deployment time for Wi-Fi is short, it provides an easy-to-manage affordable network that is private, with low latency and considerable bandwidth – without the need for an external network operator. Unfortunately, the suitability of Wi-Fi is limited. First of all, Wi-Fi offers a few non-overlapping channels, but most of its channels overlap and create interference. The presence of radio noise, which is common in production environments, further degrades the performance of a Wi-Fi network, as it operates in unlicensed spectrum. Unmanned ground vehicles (UGVs) and other types of mobile robots that move in and out of the coverage of a Wi-Fi hotspot could suffer severe impairments, like application restart, due to the time needed to reestablish the connection – as Wi-Fi does not support fast handover. The handover issue can be overcome by enabling robots to simultaneously receive more Wi-Fi channels, but with a consequent increase in cost and complexity.

LTE guarantees high bandwidth and reasonable latency, and offers full control of interference and handover. As such, LTE immediately resolves all of the issues presented by Wi-Wi. However, a standard LTE deployment requires operator involvement and a SIM for each connected terminal. Each deployment requires an agreement with an operator, incurring operational costs, which can be significant especially when it comes to international enterprises that have operations in multiple countries, necessitating several agreements.

To overcome these limitations, LTE- unlicensed, operating in a free 5GHz bandwidth, will be introduced. Without the need for an operator, this technology supports the deployment of a private LTE network, similar to Wi-Fi, but with most of the benefits of LTE. The use of unlicensed spectrum, however, may give rise to interference with other nearby radios, or from radios using the same or adjacent spectrum.

5G will be the technology of choice for applications that require very high capacity as well as very low latency. 5G networks, incorporating LTE access along with new air interfaces, will offer the coverage, bandwidth (1Gbps per user), and sub-1ms latency needed to support the time-critical applications of industries like health care and agriculture. And similar to LTE, solutions will be deployed in either licensed or unlicensed spectrum with the same benefits and drawbacks.

Benefits to industry

In manufacturing, mobile robots can be used to advantage to transport goods between various stations in a process or to and from depots. Deploying mobile robots in logistics improves productivity and supports the implementation of effective lean manufacturing. As long as there are no constraints imposed in their movement capabilities caused by unexpected obstacles or dirt, robots can carry out any sequence of events to ensure that materials arrive at the right place just in time.

In the health care sector, robots can be used to transport specimens, drugs, and bedlinen to wards, labs, pharmacies, and depositories – offloading repetitive low-level tasks from skilled hospital staff. As in the manufacturing case, the flexibility offered by mobile robots, which can be reinstructed on the fly to carry out tasks depending on contingent needs, facilitates operation optimization and cost reductions.

In agriculture, robotics can be applied to the movement of goods and equipment between fields, stalls, and barns. Mobile robots can be deployed directly in pastures and on arable land to carry out tasks like monitoring, spraying, pruning, and harvesting. Overall, the use of robotics will increase productivity and reduce opex.

At the lowest level, it is the level of performance offered by 4G, 5G, and Wi-Fi solutions in terms of bandwidth and latency that will enable robot control functionalities to be transferred to the cloud. 4G and 5G solutions are more advantageous than Wi-Fi, as they are less sensitive to interference caused by infrastructure and other machines, and provide seamless handover, ensuring continuous connectivity for robots as they move between cells.

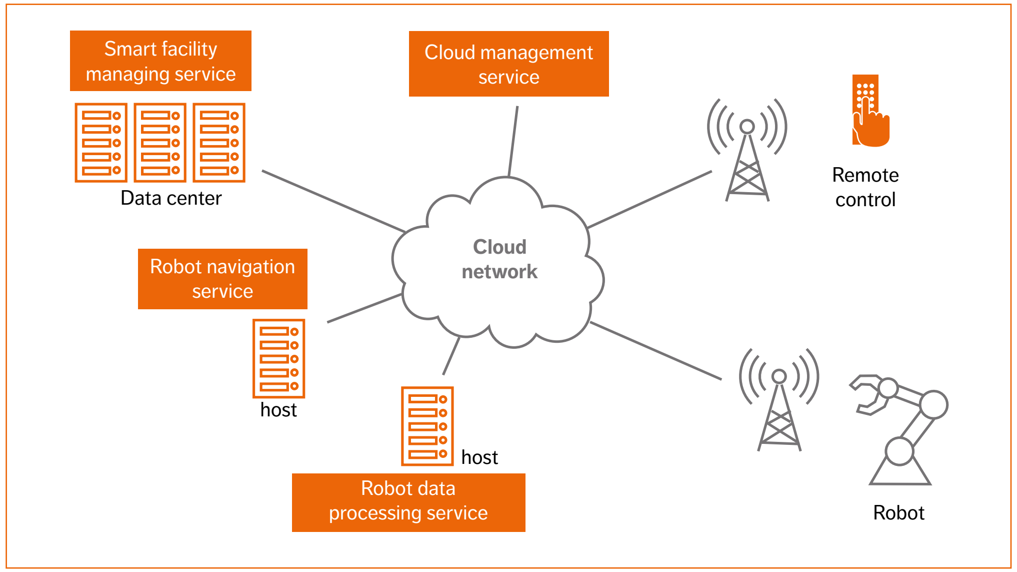

Figure 2
An example cloud architecture

The primary link characteristic that determines the stability of robot control is latency, and so for most use cases, 4G is a good starting point to provide connectivity. But as 5G supports a wider range of requirements including sub-millisecond latency, it will fulfill both the bandwidth and latency needs that applications like remote-control robotics demand. Sub-millisecond latency is needed when touch or force (haptic control) sensors are used to maintain control over robot movement. Scenarios that require visual feedback need a round-trip latency of up to 10ms – depending on the task being performed – whereas round-trip times of up to 100ms are good enough to maintain control of a robot as it moves from one place to another. Such response times imply high bandwidth availability, even though many remote applications – with the exception of those that rely on control with HD cameras and stereoscopics – often need to transfer only small amounts of data in order to function. Transferring sensor and command data usually requires several tens to a 100kbps.

Cloud intelligence

Ideally, smarts robots need high-performance, low-cost computing capabilities. To satisfy this need, cloud robotics design is based on the use of automatons with a minimal set of controls, sensors, and actuators, with system intelligence placed in the cloud, where computing capacity is unlimited. And while proper radio connectivity can satisfy latency and bandwidth demands, care needs to be taken with distribution and management of services. As shown in *Figure 2*, remote-control use cases should be distributed among a few remote cloud data centers, each dedicated to a specific task – all under the control of a cloud management service that handles network configuration, as well as the allocation of processes and VMs.

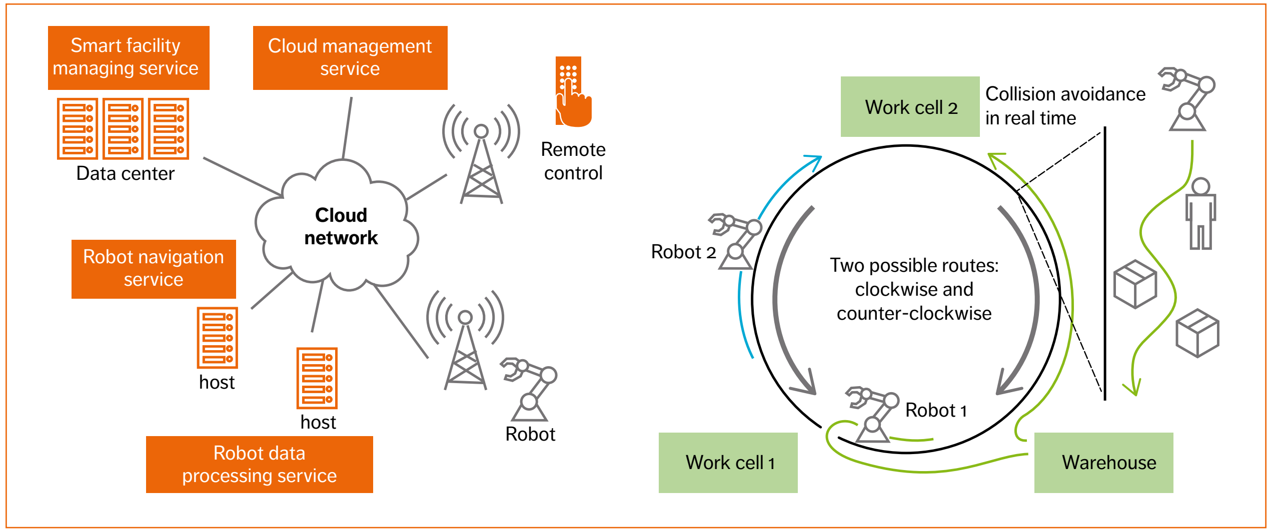

Figure 3
Logistics in a warehouse

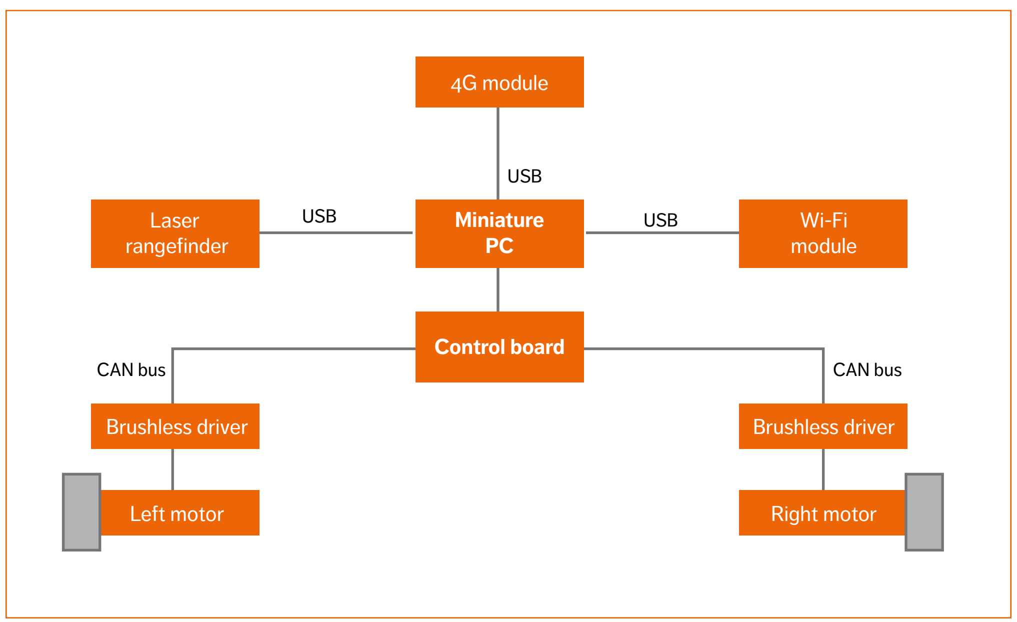

Figure 4
UGV architecture

Control services can be categorized either as time critical (such as navigation and sensor information processing) or not. To minimize the effect on latency, the cloud management service should place time-critical services close to the base station serving the working area, as doing so guarantees stability and reaction times, enabling tasks to be completed within the allocated interval. Services that are not time-critical — such as the facility management function controlling the plant — can be placed remotely, as their actions do not affect real-time behavior.

The main services required by cloud robotics are:

- smart facility management — including data analytics and robot coordination management
- image processing — for pattern recognition
- navigation

The smart facility management service coordinates user requests for services and robot activities by matching robots to requested tasks according to an optimization criterion such as minimum path length. The image-processing service involves the extraction of relevant camera data to aid navigation and other tasks. The navigation service calculates the path the robot should take to reach its destination on the basis of its current position and sensor data. It is also responsible for the collision avoidance mechanism.

To ensure robot stability, navigation tasks typically require a round-trip latency under 100ms [3], and most of this time is taken up by processing in network routers. The control loops for scenarios that use visual, force, or touch feedback are highly time critical, requiring round-trip times of less than 1ms and up to about 10ms, for some applications. To maintain robot stability in such scenarios, sensors, processing and control services should be located close to the base station providing connectivity.

In a number of cases, robot control services need to be distributed among data centers. This is the case, for example, when a robot with strict timing constraints operates in a large space like a field or a depot; or when robots move between radio cells; or for load-balancing reasons, when several mobile robots converge on the same area. In such cases, the cloud management service distributes control services in real time, transferring computational processes without disruption to services in place. This capability is typically supported by data center hypervisors. The cloud management service might in addition manage the configuration of the physical network to optimize network connectivity.

Logistics in everything

Logistics is a significant part of most industrial processes, and as such is a good use case to test the application of cloud robotics. It often involves the use of fixed and mobile robots, as well as their mutual cooperation. With this in mind, Ericsson set up a test bed — shown in the photograph in *Figure 6* — to assess the implementation of the just-in-time lean manufacturing concept, monitoring robots as they shuttled goods among warehouses and work cells, as shown in *Figure 3*.

During trials, two UGVs shuttled goods between three service areas. The UGV architecture, illustrated in *Figure 4*, includes the elements needed to drive the robot's motor, collect sensor information for low-level local control, and handle data transfer to the cloud-based remote controller.

The robots were equipped with a laser rangefinder for navigation and a webcam for pattern recognition. Trials started with a learning phase, during which the laser rangefinder built up a map of the working area — which the robot subsequently used for localization purposes. Wheel odometry controlled the robot's movements. During operation, the laser rangefinder detected obstacles in the robot's path, activating collision avoidance mechanisms when necessary.

All control services for the facility management, navigation, and data processing were located in a cloud, running in data centers or on dedicated hosts.

The navigation system used a spread middleware ROS [4]. Robot localization was implemented using odometry and 2D laser data with the AMCL [5] algorithm, while navigation was based on DWA [6] for collision avoidance and Dijkstra's algorithm for finding the shortest path.

The test area comprised a 6m x 7m rectangle, with the warehouse and work cells placed at three of the corners. Two application scenarios were tested: transportation of goods, and transportation of goods with obstacles.

Use case – transportation of goods

The first use case case – illustrated in *Figure 3* – involved a controller app and on-site workers. The robots were tasked with picking up goods at the warehouse and transporting them to one of the work cells. Pickup requests were processed by a smart facility managing service, according to the following sequence of events:

- the facility management service selects the closest available robot to the warehouse, instructing it to pick up the goods
- the controller app sends a message to a worker at the warehouse to load the robot with the goods requested
- when loading is completed, the worker uses the controller app to inform the facility management service that the robot is ready to move
- the robot proceeds to the destination along the shortest path and informs the system of its arrival
- the controller app sends a message to a worker at the destination to unload the goods from the robot
- the worker uses the controller app to inform the facility management service that the operation is complete
- the facility management service navigates the robot away from the loading bay, setting its status to available for new tasks

Use case – managing obstacles

The second use case – illustrated in *Figure 5* – aimed to reproduce a completely automated logistics process. For these trials, non-industrial robots were used. Three robotic arms were put in place to load and unload pallets from the mobile robots. An automated warehouse was simulated by a rotating platform, and two automated doors were placed along the navigation tracks. The objective of this proof-of-concept trial was to verify the interaction and coordination of multiple robots controlled by multiple remote distributed services. Each robot was assigned a dedicated control service for direct control, with the facility management service coordinating actions among the robots in real time.

Fixed and mobile obstacles and a traffic light were placed in the working area to test pattern matching and force on-the-fly rerouting on red. The laser rangefinder was used to detect obstacles, and images from the webcam were processed by cloud-based analytics to determine the status of the traffic light.

Radio connectivity was provided based on the performance characteristics of a typical public operator 4G mobile network. During trials, round-trip latency of about 40ms was measured — sufficient for stable navigation and control of robotic arms. The bandwidth available for each robot was also good enough to transfer the data collected from the camera and sensors — with a peak bitrate of 25Mbps when the reception was good, and 7Mbps when it was poor.

Latency was not a determining factor in the robots' ability to avoid obstacles or react to pattern recognition instructions. However, the presence of asymmetric NAT functionality typically used in LTE operator networks to avoid peer-to-peer communication proves to be a problem. This function can be removed but when it is in place, the robots must be designated as clients and the control services as servers, which in turn requires the robots to continuously poll the control service for updates.

Differentiation of cloud-based control processes for services that are time critical and those that are not is achieved by allocating appropriate machines to ensure correct (non-erratic) robot behavior at all times. During trials, tests were carried out on the robot's optical sensors to determine any weaknesses. The laser rangefinder was sensitive to sunlight reflections on the floor, which created fake targets that need to be removed to ensure correct navigation. The processing of images from the webcam also proved to be sensitive to light conditions, necessitating the introduction of compensation functions to properly identify the traffic light colors.

The control processes for the robotic arms were divided into three modules, two of which were cloud based. The first cloud-based process provided the correct movement sequence for the arms, in accordance with requests sent by the management service. The second cloud-based process performed

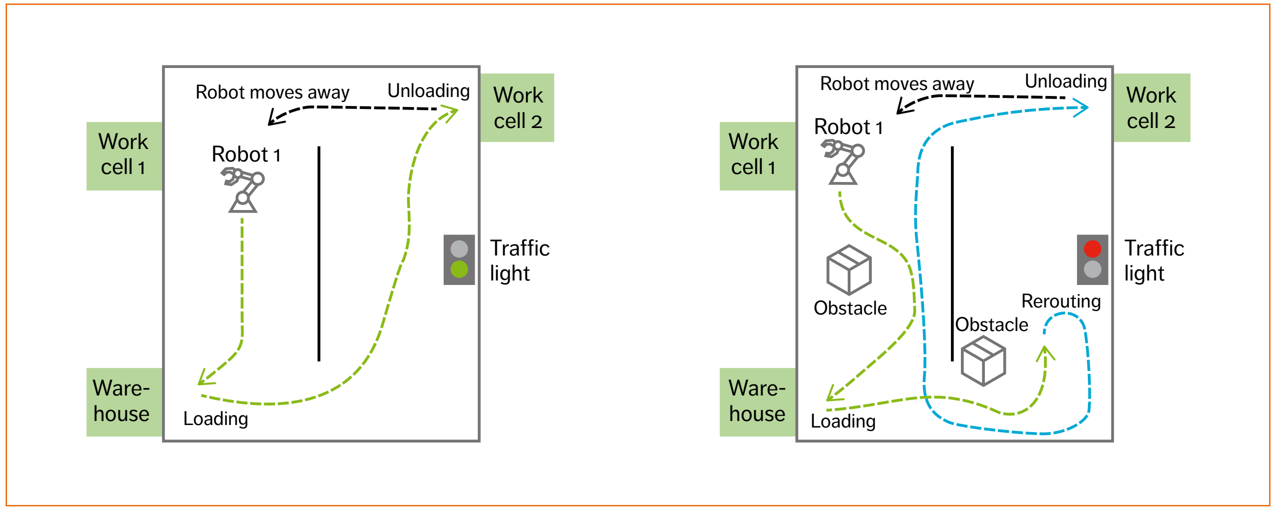

Figure 5
Tests for warehouse logistics

Figure 6
Test bed in Peccioli, Italy

the inverse kinematic transformations to translate Cartesian coordinates into joint coordinates. The third process, placed on a PC close to the robot, locally controlled the movements of the robot's joints at a low level. With round-trip latency of 40ms, the actions of the robots were stable – shorter latency was not needed as the actions involved in the trial were pick-and-place. The coordination of robots by the management service was not affected by latency in these operating conditions either. The only visible effect was a moderate slowing down of operations.

Conclusions

The interest in cloud robotics is on the rise. Its initial impact is already becoming apparent in manufacturing, agriculture, and transportation, and in scenarios where logistics can be optimized – such as harbors and hospitals – and it is likely that domestic applications will soon experience a shift in the evolution of home-help robots.

The rising level of intelligence in robots allows them to adapt to changing conditions, which is positive for development, but significantly increases their complexity. By instead connecting robots and placing this complexity (intelligence) in a cloud, affordable minimal-infrastructure smart robot systems with unlimited computing capacity will evolve. At the outset, 4G systems and Wi-Fi will be used to provide cloud robotics with the necessary connectivity, but 5G is the target technology truly capable of delivering the performance needed to support the applications of the future.

While cloud robotics may still be in its early stages of development, this article has outlined a proposed architecture for cloud robotics that relies on mobile connectivity. This architecture has been trialed, proving that the concept is viable, setting the direction for advanced cloud-based robotic systems.

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